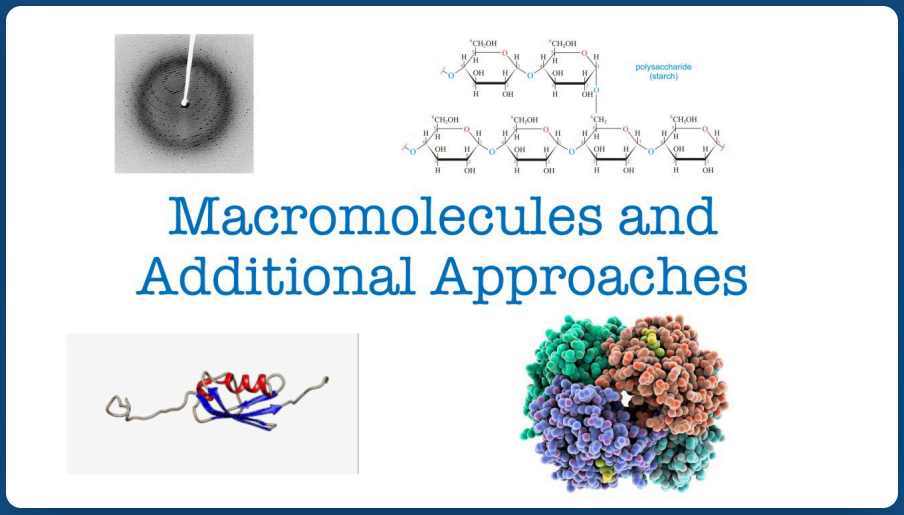
CHMD11 · 2026 · LECTURE STUDY GUIDE

Macromolecules and Additional Approaches

大分子与其他结构解析方法

中英文逐页讲解版 · Bilingual slide-by-slide guide



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slides explained
讲义页全部覆盖

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parts
个部分

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appendices: tables, worked example, self-test
个附录：总表、例题、自测

XRD · Cryo-EM · Tandem MS · Protein NMR · Nucleic Acids · Carbohydrates · Polymers

How to Use This Guide

怎么用这份讲解

Every slide of the lecture appears here as a thumbnail with an explanation in English and in Chinese. The two texts say the same thing, so read whichever is faster and use the other to check terminology. The explanations add the reasoning that a slide leaves out: why a step is needed, what a figure shows, and how the idea connects to earlier CHMD11 material.

讲义的每一页都以缩略图的形式出现在这里，配有英文和中文讲解。两种语言内容一致，读哪个快就读哪个，另一个用来核对术语。讲解补上了讲义里省略的推理：为什么需要这一步，图里画的是什么，它和 CHMD11 前面的内容怎么联系。

In one line · 一句话
The single idea to keep from the slide, with its key terms in both languages.
这一页最该记住的一个意思，附双语关键词。

要点 · Tip
Exam-oriented reminders, worked numbers and a little background beyond the slides.
面向考试的提醒、算例，以及少量讲义之外的背景。

留意 · Check
Places where the slide's wording differs from standard usage or is easy to misread. Collected in Appendix G.
讲义措辞与通行说法不一致或容易读错之处，附录 G 有汇总。

顺便一提
Short asides that make an idea easier to remember.
帮助记忆的小插曲。

说明：讲解中凡是讲义没有的内容（数值、背景、延伸）都是为了帮助理解而补充的，考试范围和标准答案以课堂与讲义为准。

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The Lecture in Ten Sentences

十句话看完这一讲

1	X-ray diffraction turns the regular spacing of atoms in a crystal into peaks at angles given by Bragg's law, 2d sin θ = nλ; it needs X-ray wavelengths and it needs a crystal.	X 射线衍射把晶体中原子的规则间距变成特定角度上的峰，角度由布拉格定律 2d sin θ = nλ 决定；它需要 X 射线波段的波长，也需要晶体。
2	Every part of a diffractometer (source, slits, sample position, monochromator, detector) trades intensity against resolution or removes a specific artefact.	衍射仪的每个部件（光源、狭缝、样品位置、单色器、探测器）要么在强度和分辨率之间做取舍，要么用来去掉某一种假信号。
3	Cryo-EM freezes molecules in vitreous ice, averages huge numbers of noisy projections and rebuilds a 3D map by Fourier methods, with no crystal required.	冷冻电镜把分子冻在玻璃态冰里，对海量带噪声的投影求平均，再用傅里叶方法重建三维密度图，不需要晶体。
4	For macromolecules the building blocks are known; the task is the higher-order structure: sequence, local folding, 3D fold, assembly.	大分子的组成单元是已知的，任务是确定高级结构：序列、局部折叠、三维折叠、组装。
5	Tandem MS reads protein sequence by database matching and tells you nothing about the fold.	串联质谱靠数据库比对读出蛋白序列，对折叠方式不提供信息。
6	Protein NMR works in solution but fights low sensitivity, overlap and fast relaxation; the remedies are $^{13}C/^{15}N$ labelling, higher field and TROSY.	蛋白质 NMR 在溶液中进行，但要克服灵敏度低、谱峰重叠和弛豫快三个问题；对策是 $^{13}C/^{15}N$ 标记、更高的磁场和 TROSY。
7	The $^{1}H–^{15}N$ HSQC gives one peak per residue and is the map on which everything else is built.	$^{1}H–^{15}N$ HSQC 每个残基给一个峰，其他所有实验都以它为底图。
8	Backbone assignment pairs an experiment that sees residues i and i−1 (CBCANNH, four peaks) with one that sees only i−1 (CBCA(CO)NNH, two peaks), and walks along the chain.	主链归属把同时看到 i 和 i−1 的实验（CBCANNH，四个峰）与只看到 i−1 的实验（CBCA(CO)NNH，两个峰）配对，沿着链逐个行走。
9	Through-bond experiments give assignments; through-space NOESY gives distances and therefore the 3D fold.	通过键的实验给出归属；通过空间的 NOESY 给出距离，从而给出三维折叠。
10	Nucleic acids are sequenced (Sanger), carbohydrates are solved mainly by NMR (anomeric protons, TOCSY/COSY/ROESY), and synthetic polymers are characterized by IR and NMR for oxidation, tacticity and branching.	核酸靠测序（桑格法）读序列；糖类主要靠 NMR 解析（异头氢、TOCSY/COSY/ROESY）；合成高分子用 IR 和 NMR 表征氧化、立构规整度和支化。

PART 1

X-ray Diffraction (XRD)

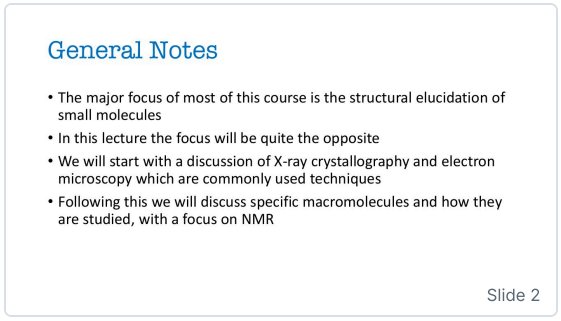
X 射线衍射

Slides 2–15

How a crystal turns an X-ray beam into a pattern of spots, and how every part of the diffractometer shapes that pattern.

晶体怎样把一束 X 射线变成一张衍射图，以及衍射仪的每个部件如何影响这张图。

SLIDE 2 General Notes 开场说明



IN ONE LINE · 一句话

The course so far: small molecules. Today: the giants.

之前学小分子，今天学大分子。

structural elucidation 结构解析　**macromolecule** 大分子

EN

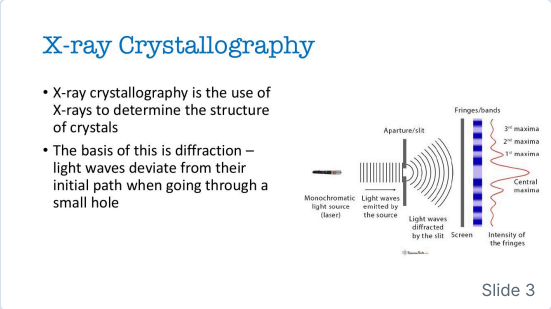
Most of CHMD11 teaches you to solve the structure of a small molecule from its spectra. This lecture flips the scale. It opens with two non-NMR techniques that dominate structural biology, X-ray crystallography and electron microscopy, and then tours the main classes of macromolecules (proteins, nucleic acids, carbohydrates, synthetic polymers), always asking: which technique answers which structural question? NMR gets the most attention.

中文

CHMD11 大部分内容是「拿到谱图，推出小分子结构」。这一讲反过来看大分子。先讲两种不属于 NMR、但在结构生物学里地位极高的技术：X 射线晶体学和电子显微镜；再依次讲蛋白质、核酸、糖类、合成高分子，核心问题始终是「哪种技术回答哪一层结构问题」。重点仍然落在 NMR 上。

要点 · Tip　这一讲偏概念和比较。复习时最有用的是三张表：技术对比（附录 A）、三共振实验速查（附录 B）、主链归属流程（附录 C）。

SLIDE 3 X-ray Crystallography: Diffraction X 射线晶体学：衍射



IN ONE LINE · 一句话

Diffraction = waves bending and interfering after passing an opening comparable to their wavelength.

衍射 = 波遇到与波长相当的孔或障碍后弯折并相互干涉。

diffraction 衍射 · **interference** 干涉 · **fringes / maxima** 条纹 / 极大

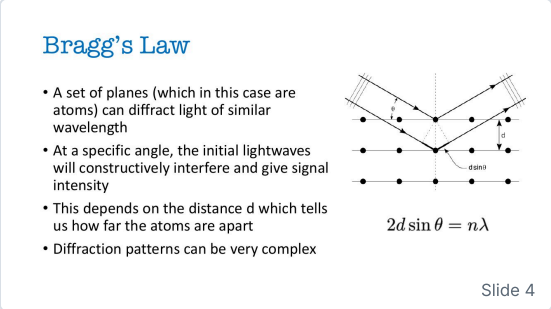
EN

X-ray crystallography determines structure from the way a crystal diffracts X-rays. The figure shows the classroom version: monochromatic light passes a slit, spreads out, and the overlapping waves add up in some directions (bright fringes, constructive interference) and cancel in others (dark, destructive). The key requirement is size matching: diffraction is strong only when the opening or spacing is comparable to the wavelength. A crystal is a three-dimensional stack of regularly spaced atoms, so it acts as a 3D diffraction grating for waves of the right wavelength.

中文

X 射线晶体学靠「晶体如何衍射 X 射线」来反推结构。图里是最经典的演示：单色光通过狭缝后散开，波与波叠加，有的方向同相叠加变亮（相长干涉），有的方向反相抵消变暗（相消干涉）。关键条件是尺寸匹配：只有当孔或间距和波长差不多大时，衍射才明显。晶体是原子在三维空间里规则排列的阵列，所以对波长合适的波来说，它就是一个三维光栅。

SLIDE 4 Bragg's Law 布拉格定律



IN ONE LINE · 一句话

2d sin θ = nλ: a peak appears only at angles where the extra path equals a whole number of wavelengths.

只有光程差等于波长整数倍的角度才出峰。

lattice planes 晶面 · **d-spacing** 晶面间距 d · **constructive interference** 相长干涉

EN

Treat the crystal as sets of parallel planes of atoms separated by a distance d. A ray reflected from the second plane travels farther than one reflected from the first; the extra path is 2d sin θ (d sin θ on the way in, d sin θ on the way out). If that extra path equals nλ, the two rays leave in phase and you detect intensity. At any other angle they cancel. So each peak position θ reports one d-spacing: measure θ, know λ, solve for d. Real crystals contain many families of planes, each with its own d, which is why patterns get complicated quickly.

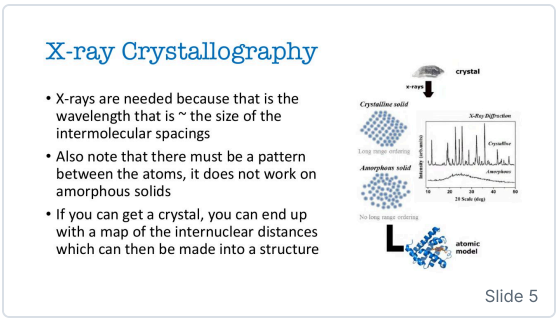
中文

把晶体看成一组组平行的原子面，面间距为 d。从第二层反射的光比第一层多走一段路，多出来的光程是 2d sin θ（进去多走 d sin θ，出来再多走 d sin θ）。当这段光程差正好等于 nλ 时，两束光同相，探测器看到强度；其他角度则相互抵消。所以每个峰的位置 θ 对应一个晶面间距：测出 θ，已知 λ，就能算出 d。真实晶体里有很多组取向不同的晶面，每组都有自己的 d，所以衍射图会很复杂。

要点 · Tip 计算题三步：① 仪器给的横轴通常是 2θ，先除以 2；② 取 n = 1；③ d = λ / (2 sin θ)。例：Cu Kα（λ = 1.5406 Å），峰在 2θ = 43.3°，则 θ = 21.65°，d = 1.5406 / (2 × 0.3689) ≈ 2.09 Å。

顺便一提 d 越小，θ 越大：原子挤得越近，峰跑得越远。倒易关系，倒着记就对了。

SLIDE 5 Why X-rays, and Why Crystals 为什么必须是 X 射线、必须是晶体



IN ONE LINE · 一句话

Wavelength must match atomic spacings (~1 Å), and the atoms must repeat periodically.

波长要匹配原子间距（约 1 Å），原子还必须周期性排列。

crystalline 晶态 · **amorphous** 非晶态 / 无定形 · **long-range order** 长程有序

EN

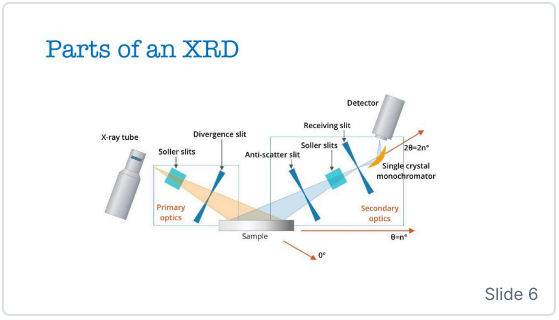
Two conditions make XRD work. First, wavelength: interatomic distances are around 1 to 3 Å, and X-rays have wavelengths in the same range, so Bragg's law has solutions at measurable angles. Visible light (thousands of Å) is far too long. Second, periodicity: sharp peaks need the same spacing repeated over thousands of unit cells. The figure contrasts a crystalline solid (long-range order, sharp peaks) with an amorphous solid (no long-range order, only a broad hump). Once you have a crystal, the pattern can be converted into a map of where the scattering matter sits, and a structure is built into that map.

中文

XRD 成立需要两个条件。第一是波长：原子间距约 1–3 Å，X 射线波长恰好在这个范围，布拉格方程才能在可测的角度上有解；可见光波长几千 Å，太长了。第二是周期性：要得到尖锐的峰，同一个间距必须在成千上万个晶胞里重复出现。图中对比了晶态固体（长程有序，尖峰）和非晶固体（无长程有序，只有一个宽包）。只要拿到晶体，就能把衍射图换算成散射物质的空间分布图，再把结构搭进去。

> **留意 · Check** 讲义写的是 “intermolecular spacings” 和 “internuclear distances”。严格说，X 射线是被电子散射的，直接得到的是电子密度图（讲义第 39 页也这么说）；原子核位置是从电子密度的中心推出来的。

SLIDE 6 Parts of an XRD: The Whole Instrument 衍射仪全貌



IN ONE LINE · 一句话

Tube → primary optics → sample → secondary optics → monochromator → detector, with detector at 2θ.

光管 → 一次光路 → 样品 → 二次光路 → 单色器 → 探测器。

goniometer 测角仪 · **θ–2θ geometry** θ–2θ 几何 · **primary / secondary optics** 一次 / 二次光学元件

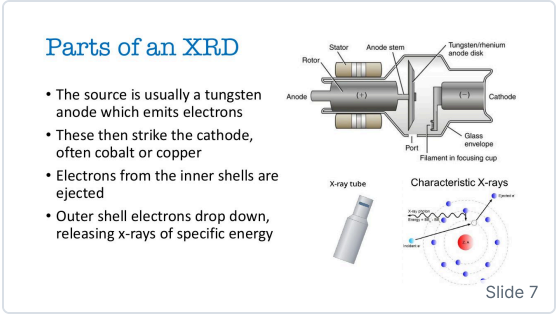
EN

This diagram is the map for the next nine slides. X-rays leave the tube, pass the primary optics (Soller slits, divergence slit), hit the flat sample at angle θ, and the diffracted beam passes the secondary optics (anti-scatter slit, Soller slits, receiving slit), a monochromator, and reaches the detector. In the common reflection geometry the detector sits at 2θ from the incident beam direction, which is why powder patterns are plotted against 2θ. A scan moves tube and detector (or sample and detector) in step so that this relationship always holds.

中文

这张图是后面九页的总地图。X 射线从光管出来，经过一次光路（索拉狭缝、发散狭缝），以角度 θ 打到平板样品上；衍射光再经过二次光路（防散射狭缝、索拉狭缝、接收狭缝）和单色器，最后进入探测器。在常用的反射几何里，探测器与入射光方向夹角为 2θ，所以粉末衍射图的横轴是 2θ。扫描时光管和探测器（或样品和探测器）联动，始终保持这个关系。

SLIDE 7 The X-ray Source X 射线源



IN ONE LINE · 一句话

Fast electrons knock out inner-shell electrons of a metal target; outer electrons fall in and emit characteristic X-rays.

高速电子打掉靶材内层电子，外层电子跃迁填补，放出特征 X 射线。

characteristic X-rays 特征 X 射线 · **Kα / Kβ** Kα / Kβ 线 · **filament** 灯丝 · **target** 靶材

EN

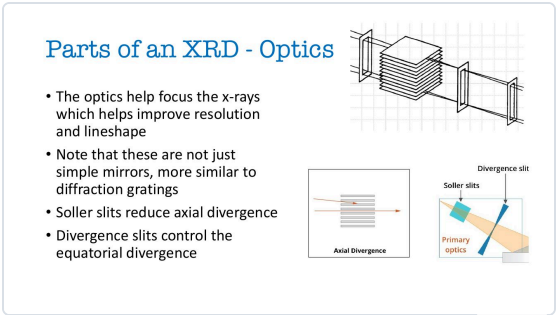
Inside the sealed tube a heated filament emits electrons, a high voltage accelerates them, and they slam into a metal target. An incoming electron ejects a core (K-shell) electron from a target atom. An electron from a higher shell drops into the hole and the energy difference leaves as an X-ray photon. Because shell energies are fixed for each element, the emitted wavelength is characteristic of the target metal: Cu Kα is 1.5406 Å, Co Kα about 1.79 Å. An L→K transition gives Kα; M→K gives the weaker, shorter-wavelength Kβ. Most of the electron energy becomes heat, which is why targets are water-cooled or rotated.

中文

密封光管里，加热的灯丝发射电子，高压把电子加速后轰击金属靶。入射电子把靶原子的内层（K 层）电子打出去，外层电子落下来填补空位，能量差以 X 射线光子的形式放出。每种元素的壳层能量是固定的，所以放出的波长是靶材的「指纹」：Cu Kα 为 1.5406 Å，Co Kα 约 1.79 Å。L→K 跃迁给出 Kα，M→K 跃迁给出更弱、波长更短的 Kβ。电子的大部分能量变成了热，所以靶需要水冷或旋转。

留意 · Check 讲义文字写的是 “tungsten anode emits electrons … strike the cathode, often cobalt or copper”，电极名称与通常的定义相反。标准说法：发射电子的是阴极（钨灯丝，cathode），被轰击并发出 X 射线的是阳极靶（Cu、Co、Mo 等，anode）。同一页的示意图里 anode disk 也是被轰击的那一方。考试若照讲义原句出题，建议向老师确认以哪个说法为准。

SLIDE 8 Primary Optics 一次光路



IN ONE LINE · 一句话

Slits shape the beam: Soller slits limit axial divergence, the divergence slit sets the equatorial spread.

狭缝给光束整形：索拉狭缝限制轴向发散，发散狭缝控制赤道方向的张角。

Soller slits 索拉狭缝 · **divergence slit** 发散狭缝 · **axial / equatorial divergence** 轴向 / 赤道发散 · **lineshape** 峰形

EN

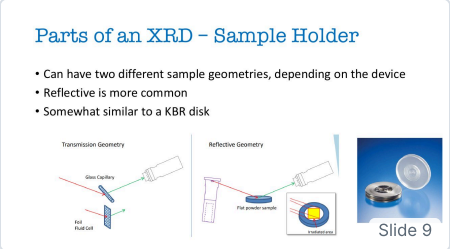
X-rays leave the tube in all directions, but Bragg's law assumes one well-defined incident angle. The primary optics narrow the beam so that every ray meets the sample at nearly the same θ. Soller slits are a stack of thin parallel metal plates that pass only rays travelling nearly parallel to the plates, which removes axial divergence (spread along the goniometer axis) and with it the asymmetric tails on low-angle peaks. The divergence slit sets how wide the beam fans out in the diffraction plane and

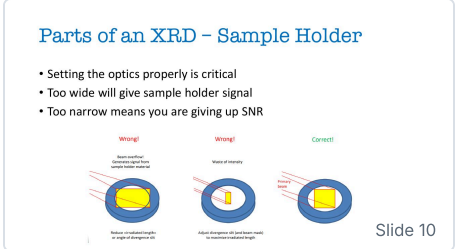
中文

X 射线从光管出来时朝四面八方发散，而布拉格定律默认入射角是一个确定的值。一次光路的任务就是把光束收窄，让所有光线几乎以同一个 θ 到达样品。索拉狭缝是一叠平行的薄金属片，只让几乎平行于金属片的光通过，消除轴向发散（沿测角仪轴方向的发散），低角度峰的不对称拖尾也随之减轻。发散狭缝决定光束在衍射平面内张开多大，也就决定了样品被照亮的面积。X 射线没法用普通镜子或透镜聚焦，所以整形靠的是狭缝、晶体和多层膜，这些元件本身就是利用衍射工作的。

therefore how much of the sample is illuminated. X-rays cannot be focused with ordinary mirrors or lenses, so beam conditioning relies on slits, crystals and multilayer devices that themselves work by diffraction.

SLIDES 9–10 Sample Holder and Beam Footprint 样品架与照射范围





IN ONE LINE · 一句话

Light only the sample: too wide a beam adds holder signal, too narrow wastes intensity.

只照样品：光太宽会打到样品架，太窄又浪费信号。

reflection geometry 反射几何
transmission geometry 透射几何
SNR 信噪比
beam overflow 光束溢出

EN

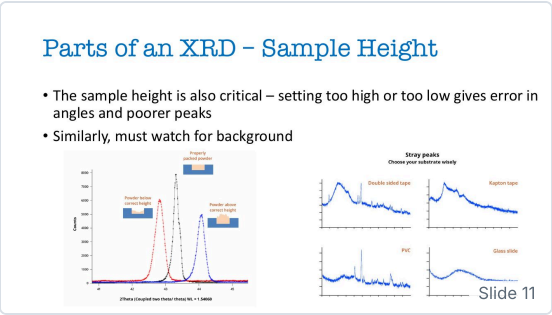
Two sample geometries exist. In reflection geometry, the common one, powder is pressed flat into a shallow well and the beam reflects off the surface. In transmission geometry the beam passes through a capillary or thin film. The slide compares the pressed powder to a KBr disk from IR: a flat, uniform, randomly oriented specimen. The second slide is about matching beam to sample. If the divergence slit is too open, the beam footprint spills over the powder and onto the holder, and the holder material adds its own peaks. If it is too closed, only a strip of the sample is used and counts are thrown away. The correct setting just fills the sample.

中文

样品几何有两种。常用的是反射几何：把粉末压平在浅槽里，光束在表面反射。透射几何则让光束穿过毛细管或薄膜。讲义把压好的粉末类比成红外里的 KBr 压片：平整、均匀、取向随机。第二页讲光束和样品的匹配。发散狭缝开得太大，光斑会超出粉末照到样品架上，样品架材料自己的峰就混进来了；开得太小，只用到样品的一条窄带，白白丢掉计数。正确的设置是光斑刚好铺满样品。

> **要点 · Tip** 注意低角度时光斑会被拉长（照射长度 ∝ 1/sin θ），所以同一个狭缝在低角度更容易溢出。

SLIDE 11 Sample Height and Background 样品高度与背景



IN ONE LINE · 一句话

A surface off the goniometer axis shifts every peak; the substrate you choose adds its own background.

样品表面偏离测角仪轴心，所有峰都会偏移；衬底材料也会贡献背景。

sample displacement error 样品位移误差
background 背景
substrate 衬底

EN

The geometry assumes the sample surface lies exactly on the goniometer axis. In the left plot the same peak is measured three times: properly packed powder (black, about 43.3°), powder below the correct height (red, shifted to lower 2θ) and powder above it (blue, shifted to higher 2θ). The misplaced

中文

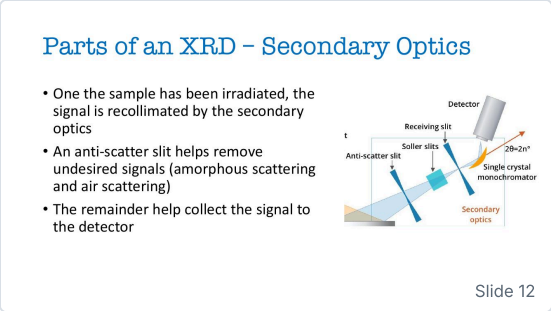
衍射几何假定样品表面正好落在测角仪的轴心上。左图把同一个峰测了三次：填装正确（黑色，约 43.3°）、样品面偏低（红色，移向低 2θ）、样品面偏高（蓝色，移向高 2θ）。位置不对的样品峰还会更矮更宽。d 是由 θ 算出来的，所以高度误差会变成所有 d 值的系

samples also give weaker, broader peaks. Since d is calculated from θ, a height error becomes a systematic error in every d-spacing. The right panels show blank scans of common mounting materials: double-sided tape and Kapton give broad humps, PVC gives sharp stray peaks, a glass slide gives the smooth hump typical of an amorphous solid. Run a blank so you know which features are not your sample.

统误差。右边四张小图是常见固定材料的空白扫描：双面胶和 Kapton 胶带有宽包，PVC 有尖锐的杂峰，玻璃片是非晶固体典型的平滑宽包。先扫空白，才知道哪些信号不属于样品。

> **顺便一提** 「样品压平」听起来像废话，但它是粉末衍射里最常见的系统误差来源。

SLIDE 12 Secondary Optics 二次光路



IN ONE LINE · 一句话

After the sample, slits re-collimate the diffracted beam and block scatter that is not Bragg diffraction.

样品之后的狭缝重新准直衍射光，并挡掉不属于布拉格衍射的散射。

anti-scatter slit 防散射狭缝 · **receiving slit** 接收狭缝 · **air scatter** 空气散射 · **collimate** 准直

EN

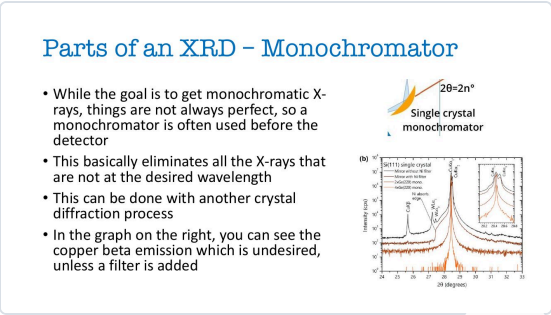
The optics on the detector side mirror those on the source side. The anti-scatter slit lets the detector see only the illuminated sample area, rejecting X-rays scattered by air along the beam path and diffuse scattering from amorphous material. A second set of Soller slits again limits axial divergence. The receiving slit defines the angular acceptance of a point detector: a narrower slit gives sharper peaks but fewer counts. This resolution versus intensity trade-off appears at every slit in the instrument.

中文

探测器一侧的光路和光源一侧是对称的。防散射狭缝让探测器只「看见」被照亮的样品区域，挡掉光路上空气的散射和非晶物质的漫散射。第二组索拉狭缝再次限制轴向发散。接收狭缝决定点探测器接收的角度范围：狭缝越窄，峰越尖，但计数越少。「分辨率换强度」这个取舍在仪器的每一道狭缝上都会出现。

> **留意 · Check** 讲义首句 “One the sample…” 应为 “Once the sample…”，笔误，不影响理解。

SLIDE 13 Monochromator 单色器



IN ONE LINE · 一句话

A crystal set at the Bragg angle for Kα passes only that wavelength and removes Kβ and other lines.

把一块晶体摆在 Kα 的布拉格角上，只让这个波长通过，滤掉 Kβ 等杂线。

monochromatic 单色的 · **Kβ filter** Kβ 滤片 · **single-crystal monochromator** 单晶单色器

EN

The tube emits Kα, Kβ and a continuous background. A second wavelength means every d-spacing produces a second peak at a different angle, which looks like extra phases. A monochromator solves this with Bragg's law itself: a single

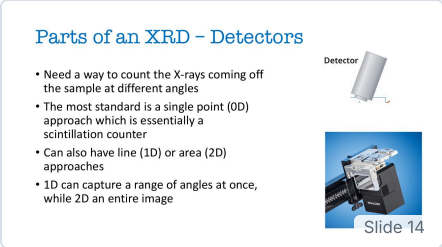
中文

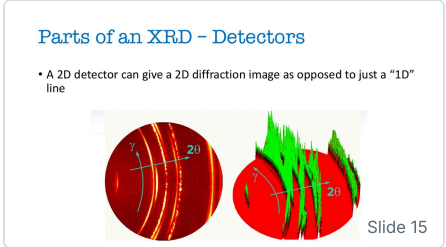
光管发出的有 Kα、Kβ，还有连续谱背景。多一个波长，每个晶面间距就会在另一个角度多出一个峰，看起来像多了别的物相。单色器用布拉格定律本身解决这个问题：把一块 d 已知的单晶固定在只对 Kα 满足 2d sin θ = λ 的角度上，其他波长就不会被反射到探测

crystal of known d is fixed at the angle that satisfies 2d sin θ = λ for Kα only, so other wavelengths are not reflected toward the detector. In the plot, the scans without filtering show a Cu Kβ satellite beside the main Si(111) reflection (and tungsten lines from an aged tube); they disappear when a filter or monochromator is used. The cheaper alternative is a thin metal foil filter, nickel for a copper tube, whose absorption edge lies between Kβ and Kα so it absorbs Kβ much more strongly.

器。图中未滤波的曲线在 Si(111) 主峰旁边能看到 Cu Kβ 的伴峰（以及光管老化带来的钨线），加了滤片或单色器之后消失。更便宜的办法是金属箔滤片：铜靶配镍箔，镍的吸收边正好在 Kβ 和 Kα 之间，对 Kβ 的吸收强得多。

SLIDES 14–15 Detectors: 0D, 1D, 2D 探测器：点、线、面





IN ONE LINE · 一句话

0D counts one angle at a time; 1D records a range of angles at once; 2D records the whole image.

点探测器一次测一个角度；线探测器一次测一段角度；面探测器直接拍整幅图。

- **scintillation counter** 闪烁计数器
- **point / line / area detector** 点 / 线 / 面探测器
- **Debye–Scherrer rings** 德拜–谢乐环

EN

A detector counts X-ray photons as a function of angle. The classic point (0D) detector is a scintillation counter: a crystal converts each X-ray photon to a flash of visible light and a photomultiplier counts the flashes. It sees one angle at a time, so a scan is slow. A line (1D) detector has many channels along 2θ and collects a range of angles simultaneously, which shortens a scan from hours to minutes. An area (2D) detector records an image. For a powder the image contains rings, because randomly oriented crystallites diffract into cones; integrating around each ring gives the familiar 1D pattern, as the second slide's figure shows. A 2D image also reveals what a line scan hides: spotty rings mean too few, too large grains, and uneven intensity around a ring means preferred orientation.

中文

探测器的任务是记录不同角度上的 X 射线光子数。经典的点探测器（0D）是闪烁计数器：晶体把每个 X 射线光子转成一次可见光闪烁，光电倍增管负责计数。它一次只看一个角度，扫描很慢。线探测器（1D）沿 2θ 方向有很多通道，可以同时采一段角度，把几小时的扫描缩短到几分钟。面探测器（2D）直接记录图像。粉末样品的图像是一圈圈的环，因为取向随机的小晶粒把光衍射成圆锥；沿每个环积分，就得到常见的一维图谱，第二页的图画的正是这个过程。二维图还能看出线扫描看不到的信息：环上有斑点说明晶粒太少太粗，环上强度不均匀说明有择优取向。

PART 2

Electron Microscopy and Cryo-EM

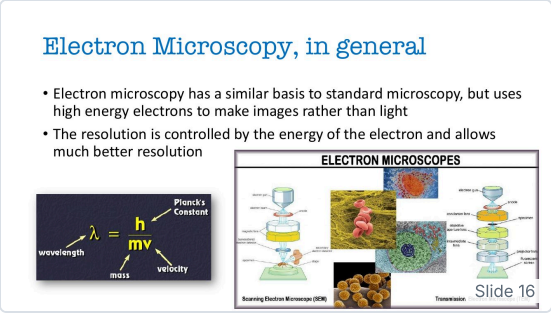
电子显微镜与冷冻电镜

Slides 16–25

Images instead of diffraction patterns: freeze the molecules, photograph millions of them, and compute a 3D map.

不看衍射图而直接看像：把分子冻住，拍下几百万个，再算出三维密度图。

SLIDE 16 Electron Microscopy in General 电子显微镜概述



IN ONE LINE · 一句话

Electrons have a far shorter wavelength than light, so they can resolve far smaller detail.

电子的波长比光短得多，所以能分辨更小的细节。

de Broglie wavelength 德布罗意波长 · **SEM** 扫描电镜 · **TEM** 透射电镜 · **resolution** 分辨率

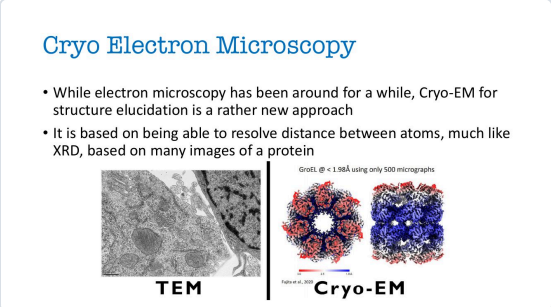
EN

A microscope cannot resolve detail much finer than the wavelength it uses, which limits light microscopes to about 200 nm. Electrons behave as waves with λ = h/mv (the formula on the slide): the faster the electron, the shorter the wavelength. At 300 kV, λ is about 0.02 Å, far smaller than an atom, so the practical limit comes from lenses and sample damage and not from wavelength. The diagram shows the two families. A scanning electron microscope (SEM) scans a beam across a surface and collects scattered electrons, giving surface images. A transmission electron microscope (TEM) sends electrons through a thin specimen and uses magnetic lenses to form an image, like a light microscope built with electromagnets. Cryo-EM is a form of TEM.

中文

显微镜分辨不了比所用波长小太多的细节，所以光学显微镜的极限在 200 nm 左右。电子也是波，λ = h/mv（讲义上的公式）：电子越快，波长越短。300 kV 下 λ 约 0.02 Å，比原子小得多，所以实际分辨率受限于透镜和样品损伤，而不是波长。示意图画了两类电镜：扫描电镜（SEM）让电子束在样品表面逐点扫描，收集散射电子，得到表面形貌；透射电镜（TEM）让电子穿过很薄的样品，用磁透镜成像，相当于一台用电磁铁做透镜的光学显微镜。冷冻电镜属于 TEM。

SLIDE 17 Cryo-EM as a Structure Method 把冷冻电镜当作结构解析手段



Slide 17

IN ONE LINE · 一句话

Like XRD it resolves interatomic distances, but from many images of single particles, with no crystal.

和 XRD 一样能分辨原子间距，但用的是大量单颗粒的图像，不需要晶体。

single-particle analysis (SPA) 单颗粒分析 · micrograph 显微照片 · near-atomic resolution 近原子分辨率

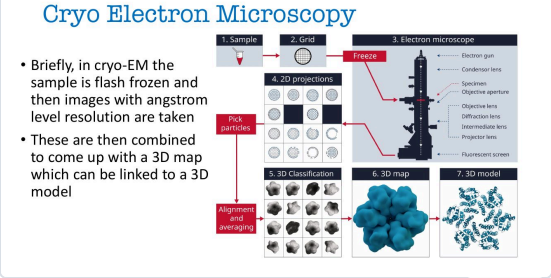
EN

Electron microscopes are old; using them to solve biomolecular structures at near-atomic resolution is recent, made possible by direct electron detectors and better image-processing software (the 2017 Nobel Prize in Chemistry recognized the method). The slide's left image is a conventional TEM picture of a stained cell section: informative about cell architecture but nowhere near atomic detail. The right image is GroEL, a large chaperone complex, reconstructed to 1.98 Å from only 500 micrographs. The difference is the strategy: instead of one image of one object, cryo-EM combines an enormous number of images of identical molecules.

中文

电镜本身历史很久，但用它把生物大分子解到近原子分辨率是近些年的事，靠的是直接电子探测器和更好的图像处理算法（2017 年诺贝尔化学奖授予了这项方法）。讲义左图是传统 TEM 拍的染色细胞切片，能看清细胞结构，但离原子细节很远；右图是大型分子伴侣 GroEL，只用了 500 张显微照片就重构到 1.98 Å。区别在于策略：不是给一个物体拍一张照，而是把海量的、彼此相同的分子的图像合并起来。

SLIDE 18 The Cryo-EM Workflow 冷冻电镜流程



Slide 18

IN ONE LINE · 一句话

Sample → grid → freeze → image → pick particles → 2D classes → 3D map → atomic model.

样品 → 载网 → 冷冻 → 成像 → 挑颗粒 → 二维分类 → 三维图 → 原子模型。

grid 载网 · particle picking 颗粒挑选 · 3D classification 三维分类 · density map 密度图

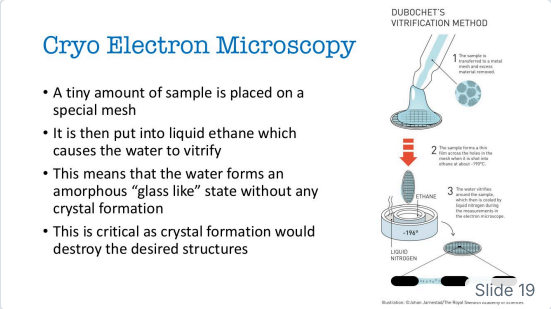
EN

The flow chart on the slide is the whole method in seven boxes. (1) Purified sample in solution. (2) A few microlitres go onto a metal grid covered with a holey support film. (3) The grid is plunge-frozen and imaged in the microscope. (4) Each micrograph contains many 2D projections of the molecule in random orientations. (5) Particles are picked, aligned, averaged and sorted, in 2D and then in 3D. (6) The result is a 3D map. (7) An atomic model is built into the map, exactly as a model is built into electron density in crystallography. The next five slides expand steps 3 to 6.

中文

讲义上的流程图用七个框概括了整个方法：（1）溶液里的纯化样品；（2）取几微升滴到带多孔支持膜的金属载网上；（3）载网快速冷冻后放进电镜成像；（4）每张照片里都有许多取向随机的分子二维投影；（5）挑出颗粒，对齐、平均、分类，先二维后三维；（6）得到三维密度图；（7）往图里搭原子模型，做法和晶体学里往电子密度里搭模型一样。后面五页是对第 3 到第 6 步的展开。

SLIDE 19 Vitrification 玻璃化冷冻



IN ONE LINE · 一句话

Plunge into liquid ethane: water becomes glass, not ice crystals.

插入液态乙烷：水变成玻璃态，而不是冰晶。

vitrification / vitreous ice 玻璃化 / 玻璃态冰 · **plunge freezing** 投入式冷冻 · **liquid ethane** 液态乙烷

EN

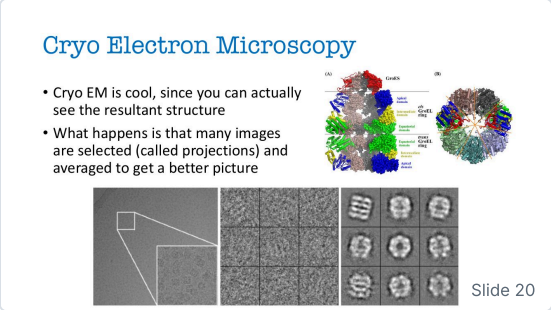
A drop of sample is applied to the grid and blotted so that only a film tens of nanometres thick spans each hole. The grid is then plunged into liquid ethane held near −180 °C by a liquid nitrogen bath. Cooling is so fast that water molecules cannot rearrange into an ice lattice and are trapped in a disordered, glass-like state: vitreous ice. This matters for two reasons. Growing ice crystals expand and would push the protein around and damage it, and crystalline ice diffracts electrons strongly, which would bury the weak protein signal. Vitreous ice keeps the molecule hydrated in a near-native state and is transparent enough to image through.

中文

把一滴样品加到载网上，用滤纸吸走多余液体，每个孔里只留下几十纳米厚的液膜，然后迅速插入由液氮浴保持在约 −180 °C 的液态乙烷。降温快到水分子来不及排成冰的晶格，被「冻」在无序的类玻璃状态，这就是玻璃态冰。这样做有两个原因：冰晶生长时体积膨胀，会挤压、破坏蛋白；晶态冰对电子的衍射很强，会把蛋白微弱的信号盖住。玻璃态冰既让分子保持在接近天然的含水状态，又足够「透明」可以成像。

> **顺便一提** 为什么不直接插液氮？液氮一碰到室温物体就沸腾，气膜隔热，降温反而慢（莱顿弗罗斯特效应）。乙烷的沸点远高于它被冷却到的温度，不会起气膜，冷得快得多。

SLIDE 20 Projections and Averaging 投影与平均



IN ONE LINE · 一句话

Each particle image is a noisy 2D projection; averaging many identical views reveals the detail.

每个颗粒的图像都是带噪声的二维投影；把大量相同视角的图像平均，细节才显现。

projection 投影 · **class average** 类平均 · **radiation damage** 辐照损伤

EN

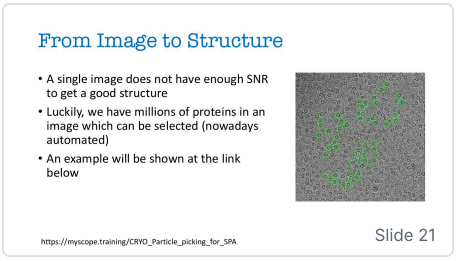
Biological molecules are destroyed by the electron beam, so each area can only receive a very low dose. The price is noise: in the raw micrograph at the bottom left of the slide, particles are faint grey smudges. Each smudge is a projection, the molecule's density summed along the beam direction, like an X-ray radiograph. Because the molecules are identical, projections taken from the same direction can be aligned and averaged. Signal adds in proportion to the number of images N while random noise grows only as √N, so signal-to-noise

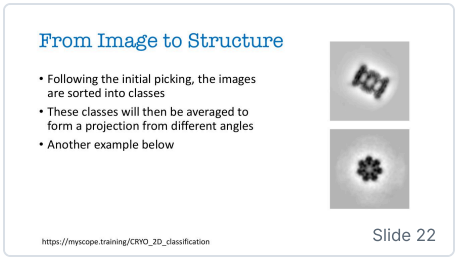
中文

生物分子会被电子束打坏，所以每个区域只能用很低的电子剂量。代价是噪声大：讲义左下角的原始照片里，颗粒只是模糊的灰点。每个灰点是一个投影，也就是分子密度沿电子束方向的叠加，像一张 X 光片。因为分子彼此相同，从同一方向拍到的投影可以对齐后求平均。信号随图像数 N 线性增加，随机噪声只按 √N 增加，所以信噪比按 √N 提高。右下角平均之后的图像里能清楚看到环和桶的形状，而原始数据里几乎什么都看不出。NMR 多次扫描累加遵循的是同一条 √N 规律。

improves as √N. The averaged images at the lower right show clear rings and barrels where the raw data showed almost nothing. The same √N rule governs signal averaging of NMR scans.

SLIDES 21–22 From Image to Structure: Picking and 2D Classification 从图像到结构：挑颗粒与二维分类





IN ONE LINE · 一句话

Find every particle, then group the look-alikes and average within each group.

先把颗粒都找出来，再把长得像的分成一组，组内求平均。

particle picking 颗粒挑选 · 2D classification 二维分类 · class 类

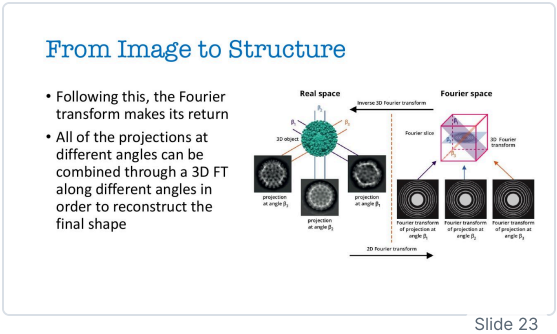
EN

One image of one molecule has too little signal for a structure, but a micrograph holds hundreds of copies and a dataset holds millions. Particle picking marks each molecule (the green circles on the slide); this used to be done by hand and is now automated with template matching or neural networks. The boxed-out particles are then sorted by 2D classification: images that look alike after rotation and shift are placed in the same class and averaged. Each class average is a clean view of the molecule from one direction; the two examples on the slide are a side view and a top view. Classification also discards junk such as ice contamination, broken particles and aggregates. The two links on the slides open interactive demonstrations of picking and classification.

中文

一个分子的一张图像信号太弱，解不出结构；但一张显微照片里有几百个拷贝，一套数据里有几百万个。颗粒挑选就是把每个分子标出来（讲义上的绿圈），以前靠手工，现在用模板匹配或神经网络自动完成。框出来的颗粒再做二维分类：经过旋转、平移后看起来相似的图像归为一类并求平均。每个类平均都是分子从某个方向看过去的清晰视图，讲义上的两张分别是侧视图和俯视图。分类还顺带剔除垃圾：冰污染、碎掉的颗粒、聚集体。两页讲义下方的链接是挑颗粒和二维分类的交互演示。

SLIDE 23 3D Reconstruction by Fourier Transform 用傅里叶变换做三维重构



IN ONE LINE · 一句话

The 2D FT of each projection is a central slice of the object's 3D FT; fill the 3D FT with slices and invert it.

每张投影的二维傅里叶变换是物体三维傅里叶变换的一个中心切片；把切片填满三维空间再反变换。

Fourier slice theorem 傅里叶切片定理 · real space / Fourier space 实空间 / 傅里叶空间 · inverse FT 傅里叶反变换

EN

The Fourier transform returns, this time in three dimensions. The figure is the projection-slice theorem. Top left: the 3D object. Below it: 2D projections at three angles. Bottom right: the 2D Fourier transform of each projection. The theorem says that each of these 2D transforms is identical to a plane through

中文

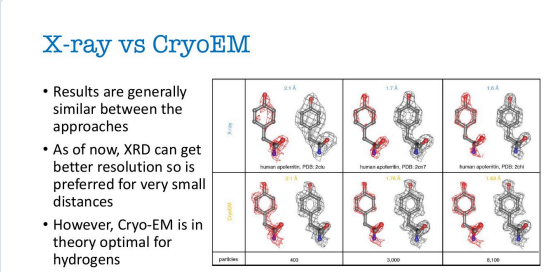
傅里叶变换再次登场，这次是三维的。图讲的是投影–切片定理。左上是三维物体，下面是三个角度的二维投影，右下是每张投影的二维傅里叶变换。定理说：每个二维变换，恰好等于物体三维傅里叶变换里过中心、且垂直于投影方向的一个平面（立方体里那个斜

the centre of the object's 3D Fourier transform, perpendicular to the projection direction (the tilted plane inside the cube). With projections from enough directions, the planes fill the cube. One inverse 3D transform then gives the object in real space. The orientations are not known in advance; software finds them iteratively by comparing each particle with projections of the current model.

面)。只要投影方向足够多，这些平面就能把立方体填满，再做一次三维反变换，就得到实空间里的物体。每个颗粒的取向事先并不知道，软件通过把颗粒与当前模型的投影反复比对、迭代求出。

要点 · Tip 类比：NMR 里 FID（时域）经 FT 变成谱（频域）；这里是投影（实空间）经 FT、拼合、反 FT 变成三维结构。CT 扫描用的是同一条定理。

SLIDE 24 X-ray vs Cryo-EM X 射线与冷冻电镜的比较



Slide 24

IN ONE LINE · 一句话

Comparable maps at comparable resolution; XRD still leads at the very high-resolution end; electrons see hydrogens better in principle.

同等分辨率下密度图相近；极高分辨率仍是 XRD 占优；电子在原理上更容易看到氢。

resolution (Å) 分辨率（埃） **electron density** 电子密度 **electrostatic potential** 静电势

EN

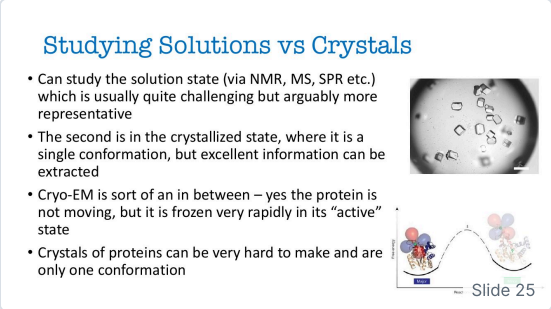
The figure compares the same residues of human apoferritin in X-ray maps (top row, 2.1, 1.7 and 1.6 Å) and cryo-EM maps (bottom row, 2.1, 1.76 and 1.63 Å). At matching resolution the mesh around the aromatic ring looks much the same: at about 2 Å the ring is a flat disc, by 1.6 Å the hole in the ring appears. The bottom line gives the number of particles used: 403, 3,000 and 8,100. More particles, better resolution. XRD remains the routine choice for the highest resolutions. The remark about hydrogens comes from the physics of scattering: X-rays scatter from electrons, and hydrogen has only one, often pulled toward its bonding partner. Electrons scatter from the electrostatic potential, to which nuclei also contribute, so hydrogen atoms are relatively more visible.

中文

图中比较了人源去铁铁蛋白同一批残基在 X 射线密度图（上排，2.1、1.7、1.6 Å）和冷冻电镜密度图（下排，2.1、1.76、1.63 Å）里的样子。分辨率相当时，芳香环周围的网格几乎一样：约 2 Å 时环是一块平板，到 1.6 Å 环中间的洞出现了。最下面一行是所用颗粒数：403、3,000、8,100，颗粒越多分辨率越高。要做最高分辨率，常规选择仍是 XRD。关于氢的那句话来自散射原理：X 射线被电子散射，而氢只有一个电子，还常被成键原子拉过去；电子被静电势散射，原子核也有贡献，所以氢原子相对更容易被看到。

要点 · Tip 「分辨率数字越小越好」：1.6 Å 比 2.1 Å 清楚。

SLIDE 25 Solution, Crystal, or Frozen? 溶液、晶体，还是冷冻?



IN ONE LINE · 一句话

Solution = realistic but hard; crystal = precise but one conformation; cryo-EM = frozen snapshots in between.

溶液态真实但难做；晶体精确但只有一种构象；冷冻电镜是介于两者之间的「定格」。

conformation 构象 · SPR 表面等离子共振 · ensemble 系综

EN

A protein in a cell is dissolved and moving. Solution methods (NMR, MS, surface plasmon resonance) observe it in that state, which is arguably the most representative, but the data are harder to obtain and interpret. A crystal gives excellent detail, yet the molecule is locked into a single conformation selected by crystal packing, and many proteins resist crystallization altogether; the photograph shows what success looks like. Cryo-EM sits between the two: the molecules are immobile, but they were frozen within milliseconds from solution, so the particles sample whatever conformations existed at that moment and 3D classification can separate them. The energy diagram on the slide makes the point: a protein occupies more than one energy well, and a method that sees only the deepest well misses part of the story.

中文

细胞里的蛋白质是溶解的、在不停运动的。溶液方法（NMR、MS、表面等离子共振）观察的就是这种状态，可以说最有代表性，但数据更难获得也更难解释。晶体能给出极好的细节，但分子被晶体堆积锁定在单一构象里，而且很多蛋白根本结不出晶体；照片展示的是结晶成功的样子。冷冻电镜介于两者之间：分子不动了，但它们是在毫秒之内从溶液里冻住的，所以颗粒保留了那一瞬间存在的各种构象，三维分类可以把它们分开。讲义上的能量曲线想说的是：蛋白质不止占据一个能量阱，只能看到最深那个阱的方法会漏掉一部分信息。

PART 3

Macromolecules and Protein Architecture

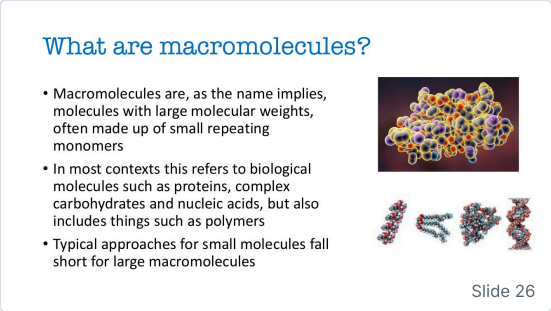
大分子与蛋白质的结构层次

Slides 26–35

What a macromolecule is, the four levels of protein structure, and why membrane proteins are so hard.

什么是大分子，蛋白质的四级结构，以及膜蛋白为什么难研究。

SLIDE 26 What Are Macromolecules? 什么是大分子?



IN ONE LINE · 一句话

Very large molecules built from repeating small units; small-molecule methods alone are not enough.

由重复小单元组成的巨大分子；只靠小分子那套方法不够用。

monomer 单体 · biopolymer 生物高分子 · molecular weight 分子量

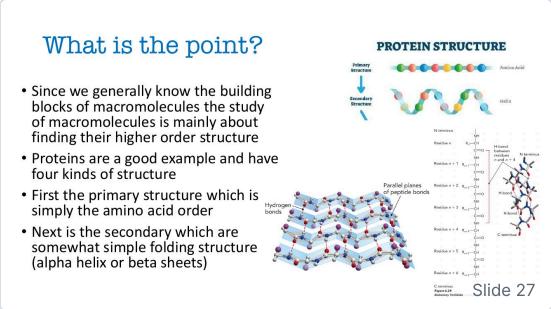
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A macromolecule has a high molecular weight and is usually assembled from small repeating monomers. In most contexts the word means biological molecules, namely proteins (from amino acids), nucleic acids (from nucleotides) and complex carbohydrates (from sugars), but synthetic polymers count too. The tools of the earlier lectures struggle here for a simple reason: a 20 kDa protein has well over a thousand hydrogens in very similar environments, so a 1D proton spectrum is an unresolved envelope, and a molecular formula from MS says almost nothing about shape.

中文

大分子的分子量很大，通常由重复的小单体连接而成。多数语境下它指生物分子：蛋白质（单体是氨基酸）、核酸（核苷酸）、复杂糖类（单糖），但合成高分子也算。前面几讲的工具在这里吃力，原因很简单：一个 20 kDa 的蛋白有一千多个化学环境非常相近的氢，一维氢谱就是一片分不开的包络；质谱给出的分子式对形状几乎没有说明力。

SLIDE 27 What Is the Point? Primary and Secondary Structure 研究目的：一级与二级结构



IN ONE LINE · 一句话

We already know the building blocks, so the real question is how the chain is ordered and folded.

组成单元是已知的，真正的问题是链的顺序和折叠方式。

primary structure 一级结构 · **secondary structure** 二级结构 · **α-helix** α 螺旋 · **β-sheet** β 折叠 · **hydrogen bond** 氢键

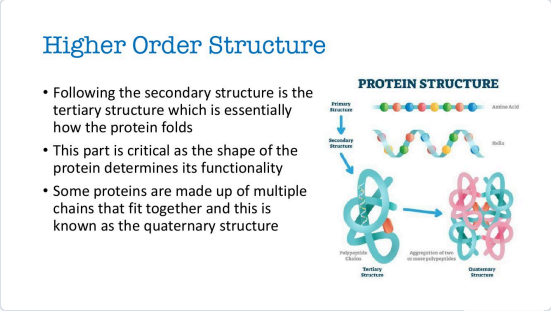
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For a small molecule, elucidation means finding the connectivity. For a protein the connectivity type is known in advance: twenty amino acids joined by peptide bonds. What is unknown is the higher-order structure. Proteins have four levels. Primary structure is the sequence of amino acids. Secondary structure is the local, regular folding of the backbone held by hydrogen bonds between backbone C=O and N–H groups: in an α-helix the C=O of residue i bonds to the N–H of residue i+4, giving a coil; in a β-sheet, extended strands lie side by side and bond to each other, in parallel or antiparallel fashion.

中文

对小分子，结构解析就是找出连接方式。对蛋白质，连接方式的类型事先就知道：二十种氨基酸靠肽键连起来。未知的是高级结构。蛋白质结构分四级。一级结构是氨基酸的排列顺序。二级结构是主链局部的规则折叠，靠主链 C=O 和 N–H 之间的氢键维持：α 螺旋里，第 i 个残基的 C=O 与第 i+4 个残基的 N–H 成氢键，形成螺旋；β 折叠里，伸展的肽段并排排列、相互成氢键，方向可以平行也可以反平行。

SLIDE 28 Tertiary and Quaternary Structure 三级与四级结构



IN ONE LINE · 一句话

Tertiary = the 3D fold of one chain, which sets function; quaternary = several folded chains assembled.

三级 = 一条链的三维折叠，决定功能；四级 = 多条折叠好的链组装在一起。

tertiary structure 三级结构 · **quaternary structure** 四级结构 · **polypeptide chain** 多肽链 · **subunit** 亚基

EN

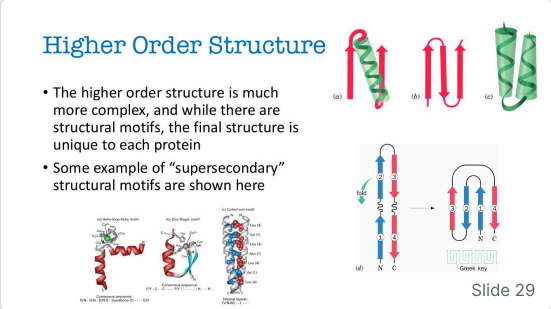
Tertiary structure is how the helices, sheets and connecting loops of one polypeptide chain pack into a three-dimensional shape. It is driven mainly by side-chain interactions: burial of hydrophobic residues, hydrogen bonds, salt bridges and disulfide bonds. This level matters most because shape determines function; an enzyme's active site exists only when the fold brings distant residues together. Some proteins consist of several chains, each with its own tertiary structure, that associate into one functional unit. That arrangement is the quaternary structure.

中文

三级结构是一条多肽链里的螺旋、折叠和连接它们的环区如何堆积成三维形状，主要由侧链间的相互作用驱动：疏水残基埋到内部、氢键、盐桥、二硫键。这一级最重要，因为形状决定功能：酶的活性位点只有在折叠把序列上相距很远的残基拉到一起之后才存在。有些蛋白由多条链组成，每条链各有自己的三级结构，再组装成一个功能单位，这种组装方式就是四级结构。

要点 · Tip 四级结构一句话对应：一级 = 顺序；二级 = 主链氢键形成的局部规则结构；三级 = 单链整体折叠；四级 = 多链组装。

SLIDE 29 Supersecondary Motifs 超二级结构模体



IN ONE LINE · 一句话

Recurring combinations of a few helices and strands, the vocabulary from which folds are built.

少数几个螺旋和 β 链的固定搭配，是搭建折叠的「词汇」。

motif 模体 · **β-hairpin** β 发夹 · **Greek key** 希腊钥匙 · **coiled coil** 卷曲螺旋 · **zinc finger** 锌指

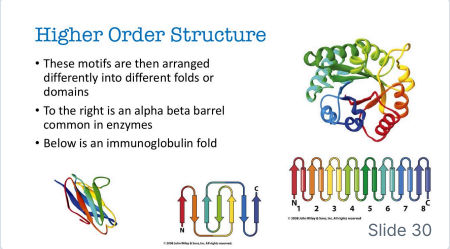
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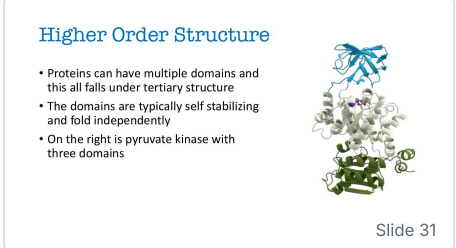
Between secondary and tertiary structure sit small recurring arrangements called motifs or supersecondary structures. The top row of the figure shows three classics: β–α–β (two parallel strands connected through a helix), the β-hairpin (two antiparallel strands joined by a tight turn) and α–α (two packed helices). The middle shows how a long hairpin folds over to give the Greek key, four antiparallel strands in a characteristic order. The bottom row shows functional motifs: helix–loop–helix, the zinc finger, where a Zn^{2+} ion holds a small domain together, and the coiled coil, two helices wound around each other. The same motifs appear in unrelated proteins, but the complete fold remains unique to each protein.

中文

在二级和三级结构之间，有一些反复出现的小型组合，叫模体或超二级结构。图上排是三种经典模体：β–α–β（两条平行 β 链经一段螺旋相连）、β 发夹（两条反平行 β 链由一个紧凑转角相连）、α–α（两条螺旋相互堆积）。中间展示一条长发夹如何对折成希腊钥匙，也就是四条反平行链按特定顺序排列。下排是带功能的模体：螺旋–环–螺旋、由 Zn^{2+} 固定的小结构域锌指，以及两条螺旋互相缠绕的卷曲螺旋。同一种模体会出现在毫不相干的蛋白里，但完整的折叠仍是每个蛋白独有的。

SLIDES 30–31 Folds and Domains 折叠类型与结构域





IN ONE LINE · 一句话

Motifs combine into folds; a fold that is stable and folds on its own is a domain.

模体组合成折叠；能独立折叠、自身稳定的折叠单位就是结构域。

fold 折叠类型 · **domain** 结构域 · **α/β barrel (TIM barrel)** α/β 桶 · **immunoglobulin fold** 免疫球蛋白折叠

EN

Motifs are arranged into larger patterns called folds. The slide shows two. The α/β barrel, common in enzymes, has eight parallel β-strands forming a closed barrel in the centre, surrounded by eight α-helices; it is eight β–α units in a ring. The immunoglobulin fold is a sandwich of two antiparallel β-sheets, the building block of antibodies. A domain is a compact region of a chain, typically 50 to 300 residues, that is stable on its own and folds independently. A protein can have several domains, often with separate jobs such as binding a substrate, binding a

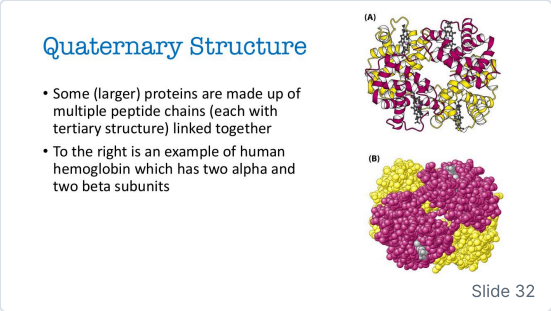
中文

模体进一步组合成更大的格局，叫折叠类型。讲义举了两个例子。α/β 桶常见于酶：八条平行 β 链在中心围成封闭的桶，外面包着八条 α 螺旋，相当于八个 β–α 单元围成一圈。免疫球蛋白折叠是两片反平行 β 折叠叠成的「三明治」，是抗体的基本构件。结构域是链上的一段紧凑区域，通常 50–300 个残基，自身稳定、可以独立折叠。一个蛋白可以有多个结构域，常常分工不同：结合底物、结合调节分子、负责催化。这些都还属于三级结构，因为仍是一条链。第二页的丙酮酸激酶用不同颜色画出了它的三个结构域。

regulator or performing catalysis. All of this is still tertiary structure, because it is one chain. Pyruvate kinase, on the second slide, is drawn with its three domains in different colours.

> **要点 · Tip** 结构域 ≠ 亚基。结构域在同一条链上；亚基是不同的链。

SLIDE 32 Quaternary Structure: Hemoglobin 四级结构：血红蛋白



IN ONE LINE · 一句话

Human hemoglobin = two α and two β chains ($\alpha_2\beta_2$), each carrying one heme.

人血红蛋白 = 两条 α 链加两条 β 链（$\alpha_2\beta_2$），每条带一个血红素。

subunit 亚基 · **tetramer 四聚体** · **heme 血红素** · **cooperativity 协同效应**

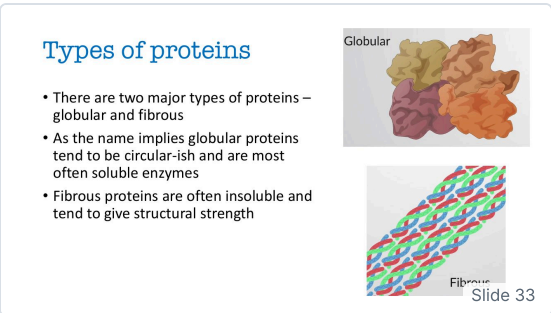
EN

Larger proteins are often built from several polypeptide chains held together mostly by non-covalent contacts. Human hemoglobin has four subunits, two α and two β (panel A ribbons, panel B space-filling), and each holds a heme group that binds one O_2. Quaternary structure brings functions no single chain has: when one subunit binds oxygen, the contacts between subunits shift and the others bind more easily. For structure determination, assemblies of this kind are large, and often symmetric, which makes them good targets for crystallography and cryo-EM and difficult ones for solution NMR.

中文

较大的蛋白常由多条多肽链组成，主要靠非共价作用结合在一起。人血红蛋白有四个亚基，两条 α、两条 β（A 图为飘带模型，B 图为空间填充模型），每个亚基带一个血红素，可结合一个 O_2。四级结构带来单条链没有的功能：一个亚基结合氧之后，亚基之间的接触发生变化，其余亚基更容易结合氧。从结构解析的角度看，这类组装体很大，而且常有对称性，适合用晶体学和冷冻电镜研究，对溶液 NMR 则很困难。

SLIDE 33 Globular and Fibrous Proteins 球状蛋白与纤维状蛋白



IN ONE LINE · 一句话

Globular: compact, soluble, usually enzymes. Fibrous: elongated, insoluble, structural.

球状：紧凑、可溶，多为酶。纤维状：细长、不溶，起结构支撑作用。

globular protein 球状蛋白 · **fibrous protein 纤维状蛋白** · **solubility 溶解性**

EN

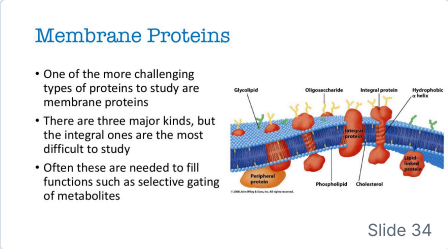
Proteins fall into two broad shape classes. Globular proteins fold into compact, roughly spherical shapes with hydrophobic side chains inside and polar ones on the surface, which makes them water-soluble; most enzymes, transport proteins and antibodies are globular. Fibrous proteins form long strands or sheets from repetitive sequences, are usually insoluble and provide mechanical strength: collagen in connective tissue, keratin in hair and nails, silk fibroin. The class predicts the

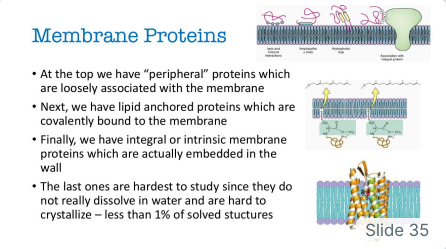
中文

蛋白质按形状大致分两类。球状蛋白折叠成紧凑、近似球形的结构，疏水侧链在内、极性侧链在表面，因此溶于水；多数酶、转运蛋白和抗体都是球状蛋白。纤维状蛋白由重复序列形成长丝或片层，通常不溶，提供机械强度：结缔组织里的胶原蛋白、头发和指甲里的角蛋白、丝素蛋白。类别决定了用什么实验：可溶的球状蛋白适合溶液 NMR 和结晶；不溶的纤维既溶不了也很少能长成单晶，要用纤维衍射、固体 NMR 或电镜。

experiment. Soluble globular proteins suit solution NMR and crystallization. Insoluble fibres do not dissolve and rarely form single crystals, so they call for fibre diffraction, solid-state NMR or electron microscopy.

SLIDES 34–35 Membrane Proteins 膜蛋白





IN ONE LINE · 一句话

Peripheral, lipid-anchored, integral. Integral ones are the hardest: they neither dissolve in water nor crystallize readily.

外周、脂锚定、整合三类。整合膜蛋白最难：不溶于水，也难结晶。

- **peripheral protein** 外周膜蛋白
- **lipid-anchored protein** 脂锚定蛋白
- **integral (intrinsic) protein** 整合膜蛋白
- **lipid bilayer** 脂双层
- **transmembrane helix** 跨膜螺旋

EN

Membrane proteins handle transport and signalling across the lipid bilayer, including the selective gating of ions and metabolites. There are three kinds. Peripheral proteins sit on the membrane surface, loosely attached by electrostatic contacts and hydrogen bonds. Lipid-anchored proteins are covalently linked to a lipid that inserts into the bilayer. Integral, or intrinsic, proteins are embedded in the membrane and usually span it, with hydrophobic transmembrane helices or a β-barrel in contact with the lipid tails. Integral proteins are the hard ones. Their hydrophobic belt makes them insoluble in water, and removing them from the membrane requires detergents or lipid nanodiscs that must imitate the bilayer without unfolding the protein. They also crystallize poorly, which is why the slide notes that they account for under 1% of solved structures, even though a large share of drug targets are membrane proteins. Cryo-EM has changed this field more than any other.

中文

膜蛋白负责跨脂双层的物质转运和信号传递，包括对离子和代谢物的选择性通过。分三类。外周膜蛋白贴在膜表面，靠静电作用和氢键松散结合。脂锚定蛋白与一个脂质分子共价相连，由脂质插入膜中。整合（内在）膜蛋白嵌在膜里，通常贯穿整个膜，由疏水的跨膜螺旋或 β 桶与脂质尾部接触。难的是整合膜蛋白：那一圈疏水表面使它们不溶于水，把它们从膜里取出来要用去垢剂或脂质纳米盘，既要模拟脂双层，又不能让蛋白变性。它们还很难结晶，所以讲义说它们在已解析的结构里占不到 1%，而药物靶点里膜蛋白的比例却很高。冷冻电镜对这个领域的改变比对其他任何领域都大。

> **留意 · Check** 讲义写 "embedded in the wall"，这里的 wall 指的是细胞膜（membrane）。细胞壁（cell wall）是另一种结构，不要混淆。

PART 4

Tandem MS and X-ray Crystallography for Proteins

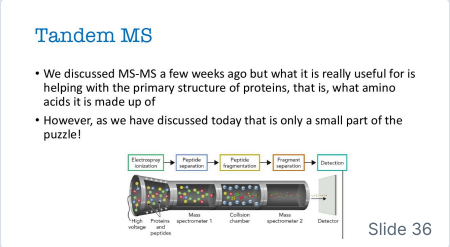
串联质谱与蛋白质晶体学

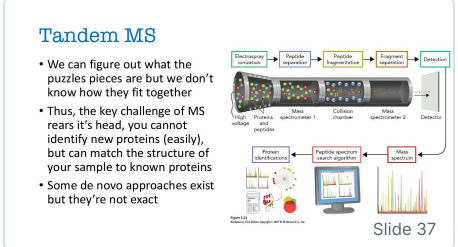
Slides 36–39

MS reads the sequence; crystallography gives the shape, if the protein will crystallize.

质谱读序列；晶体学给出形状，前提是蛋白肯结晶。

SLIDES 36–37 Tandem MS: Primary Structure Only 串联质谱：只能给出一级结构





IN ONE LINE · 一句话

MS/MS identifies the pieces and their order by matching to databases, but says nothing about the fold.

MS/MS 通过数据库比对鉴定碎片及其顺序，但不提供折叠信息。

- **tandem MS (MS/MS)** 串联质谱
- **electrospray ionization (ESI)** 电喷雾电离
- **collision cell** 碰撞室
- **database search** 数据库检索
- **de novo sequencing** 从头测序

EN

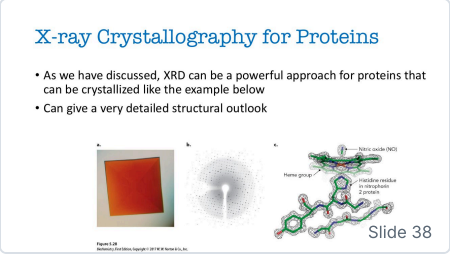
The diagram shows the standard proteomics pipeline. The protein is digested into peptides, usually with trypsin. Electrospray ionization puts the peptides into the gas phase as ions. The first mass analyser selects one peptide ion; in the collision chamber it collides with inert gas and breaks, mainly at peptide bonds; the second analyser measures the fragments. Neighbouring fragments differ by the mass of one amino acid residue, so the ladder of fragment masses spells out the sequence. In practice the fragment spectrum is compared by a search algorithm with spectra predicted from a database of known protein sequences (the lower half of the figure). This is the slide's point about the main limitation: MS is excellent at recognizing a protein that is already in a database and much weaker at characterizing an unknown one. De novo sequencing reads the ladder directly, but gaps in the fragment series and isobaric residues such as leucine and isoleucine leave ambiguities. And sequence is only the first of the four structural levels.

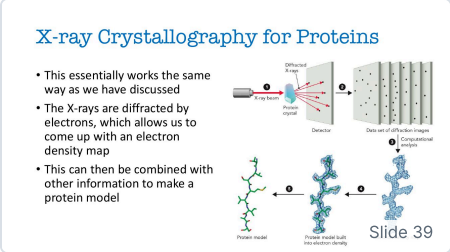
中文

图中是蛋白质组学的标准流程。蛋白先被酶切成肽段，通常用胰蛋白酶。电喷雾把肽段变成气相离子。第一级质量分析器选出一个肽段离子；它在碰撞室里与惰性气体碰撞，主要在肽键处断裂；第二级质量分析器测碎片。相邻碎片的质量差等于一个氨基酸残基的质量，所以碎片质量形成的「梯子」就能读出序列。实际操作中，检索算法把碎片谱与已知蛋白序列数据库预测出的谱图比对（图的下半部分）。这正是讲义强调的主要局限：质谱很擅长认出数据库里已有的蛋白，但鉴定未知蛋白要弱得多。从头测序直接读梯子，但碎片系列不完整，再加上亮氨酸和异亮氨酸这类同质量残基，都会留下不确定性。而且序列只是四级结构里的第一级。

> **顺便一提** 讲义的比喻很准：质谱告诉你拼图有哪些块，但盒子上的图案要靠别的方法看。

SLIDES 38–39 X-ray Crystallography for Proteins 蛋白质 X 射线晶体学





IN ONE LINE · 一句话

Crystal → diffraction spots → electron density map → model built into the density.

晶体 → 衍射点 → 电子密度图 → 把模型搭进密度里。

- **electron density map** 电子密度图
- **model building** 模型搭建
- **phase problem** 相位问题
- **refinement** 精修

EN

The principle is the same as for a salt crystal, only the unit cell is far larger and the pattern has tens of thousands of spots. The first slide shows the three stages for a heme protein: (a) the crystal, red because of the heme, (b) its diffraction image and (c) the resulting map with the heme, a bound nitric oxide and a histidine fitted into the mesh. The second slide shows the workflow. X-rays are scattered by electrons, so the computed result is a 3D electron density map, not a list of atoms. The crystallographer threads the known amino acid sequence through the density and refines the model until the diffraction pattern calculated from it agrees with the measured one. This is the "other information" on the slide: sequence, standard bond lengths and angles, and stereochemistry.

中文

原理和盐晶体一样，只是晶胞大得多，衍射图上有几万个点。第一页展示了一个血红素蛋白的三个阶段：(a) 晶体，因为含血红素而呈红色；(b) 它的衍射图；(c) 得到的密度图，血红素、结合的一氧化氮和一个组氨酸都嵌在网格里。第二页是流程。X 射线被电子散射，所以计算得到的是三维电子密度图，而不是一张原子列表。晶体学家把已知的氨基酸序列「穿」进密度里，再精修模型，直到由模型算出的衍射图与实测一致。讲义说的 “other information” 就是这些：序列、标准键长键角、立体化学。

要点 · Tip 延伸（讲义未展开）：探测器只记录衍射点的强度，记录不了相位，而计算密度图两者都需要，这就是「相位问题」，要靠分子置换或重原子等方法解决。另外氢原子在蛋白质 X 射线结构里通常看不见，这正是第 24 页说冷冻电镜「理论上更适合氢」的背景。

PART 5

Protein NMR

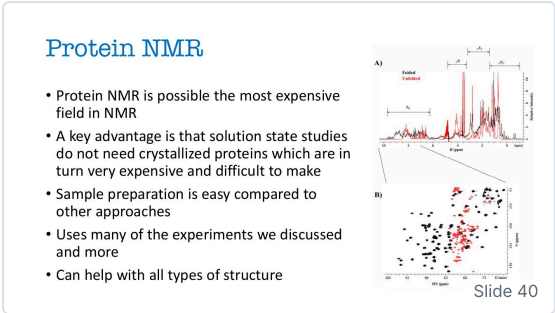
蛋白质核磁共振

Slides 40–63

The core of the lecture: why proteins are hard for NMR, how labelling and high field fix it, and how triple-resonance experiments assign the backbone one residue at a time.

本讲的核心：蛋白质为什么难做 NMR，同位素标记和高场如何解决问题，三共振实验如何一个残基一个残基地归属主链。

SLIDE 40 Protein NMR: Why Bother? 为什么要做蛋白质 NMR



IN ONE LINE · 一句话

Works in solution with no crystal, and reports on every level of structure, at a price.

在溶液里做，不需要晶体，各级结构都能给出信息，只是很贵。

solution-state NMR 溶液核磁 · **folded / unfolded** 折叠 / 去折叠 · **chemical shift dispersion** 化学位移分散度

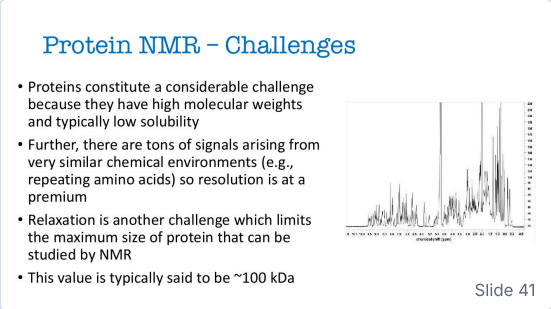
EN

Protein NMR is described on the slide as possibly the most expensive field in NMR, because of the magnets and the isotopes. The main advantage is that the protein is studied in solution, close to physiological conditions, with no crystal needed. Sample preparation is simple: dissolve the protein in buffer with a few percent D_2O for the lock. The experiments are extensions of COSY, TOCSY, HSQC and NOESY from earlier lectures, and together they report on primary through quaternary structure and on motion and binding as well. The figure shows a first quality check. A folded protein gives a 1D spectrum with widely spread peaks, including methyl signals shifted below 0.5 ppm by nearby aromatic rings and amide signals out past 9 ppm. An unfolded protein gives the same narrow, crowded spectrum as a mixture of free amino acids.

中文

讲义说蛋白质 NMR 可能是 NMR 里最烧钱的方向，贵在磁体和同位素。最大的优点是在溶液里研究蛋白，接近生理条件，不需要晶体。制样简单：把蛋白溶在缓冲液里，加百分之几的 D_2O 用于锁场。所用实验是前面学过的 COSY、TOCSY、HSQC、NOESY 的扩展，合起来能给出一级到四级结构的信息，还能看运动和结合。图里展示了第一步质检：折叠良好的蛋白，一维谱的峰分得很开，有被邻近芳环屏蔽到 0.5 ppm 以下的甲基信号，也有 9 ppm 以外的酰胺信号；去折叠的蛋白，谱图又窄又挤，和一堆游离氨基酸混在一起差不多。

SLIDE 41 Challenges: Size, Overlap, Relaxation 三大难题：分子量、重叠、弛豫



IN ONE LINE · 一句话

Big molecule → many similar signals and fast T_2 relaxation → broad, overlapping, weak peaks.

分子大 → 相似信号多、T_2 弛豫快 → 峰宽、重叠、信号弱。

transverse relaxation (T_2) 横向弛豫 · **linewidth** 线宽 · **tumbling / correlation time** 翻滚 / 相关时间 · **kDa** 千道尔顿

EN

Three problems grow with molecular weight. Sensitivity: proteins are often soluble only to about 1 mM or less, and that concentration is shared among thousands of distinct nuclei. Resolution: twenty amino acids repeat many times, so hundreds of protons have nearly identical environments; the slide's 1D spectrum is a continuous envelope. Relaxation: a large molecule tumbles slowly in solution, slow tumbling makes transverse (T_2) relaxation fast, and linewidth is proportional to $1/T_2$. Lines get broader as the protein gets bigger. Broad lines overlap more, and signal is also lost during the delays of multidimensional pulse sequences, so beyond a certain size the experiments simply stop working. The slide puts the usual limit near 100 kDa.

中文

有三个问题会随分子量增大而加重。灵敏度：蛋白往往只能溶到 1 mM 左右甚至更低，这点浓度还要分给几千种不同的核。分辨率：二十种氨基酸反复出现，几百个氢的环境几乎相同，讲义上的一维谱是一片连续的包络。弛豫：大分子在溶液里翻滚慢，翻滚慢导致横向（T_2）弛豫快，而线宽正比于 $1/T_2$，所以蛋白越大线越宽。线宽了更容易重叠，而且信号在多维脉冲序列的延迟期间也在不断损失，大到一定程度实验就做不出来了。讲义给的常见上限是约 100 kDa。

> **要点 · Tip** 补充：100 kDa 这个数字是用上了氘代和 TROSY 之后的水平。不用这些手段的常规三共振实验，大致做到 25–30 kDa 就很吃力了。

SLIDE 42 Solution 1: Isotopic Labelling 对策一：同位素标记



IN ONE LINE · 一句话

Grow the protein on ^{15}N and ^{13}C sources so every N and C is NMR-active.

用 ^{15}N、^{13}C 原料表达蛋白，让每个氮和碳都有 NMR 信号。

isotopic labelling 同位素标记 · **uniform $^{13}C/^{15}N$ labelling** 均匀 $^{13}C/^{15}N$ 标记 · **recombinant expression** 重组表达 · **natural abundance** 天然丰度

EN

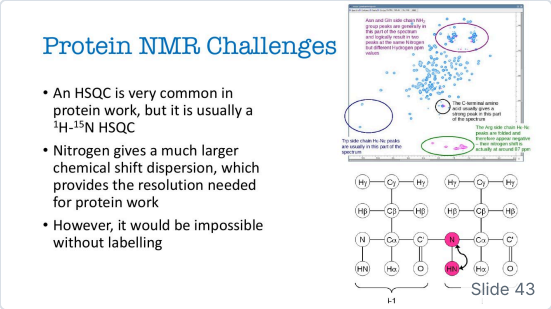
The slide says the first two problems were solved by throwing money at them, and shows the evidence: a catalogue page for $^{13}C,^{15}N$-labelled human ubiquitin at several thousand pounds. The useful isotopes are rare in nature, ^{13}C about 1.1% and ^{15}N about 0.37%, and the abundant ^{14}N has a quadrupole moment that broadens its signals beyond use. The remedy is to express the protein in bacteria, typically E. coli, grown on a minimal medium whose only nitrogen source is $^{15}NH_4Cl$ and whose only carbon source is ^{13}C-glucose. Every nitrogen and carbon in the product is then spin-½. Signal rises by roughly a hundredfold

中文

讲义说前两个难题是「用钱砸」解决的，还附了证据：^{13}C、^{15}N 标记的人泛素，商品目录价几千英镑。有用的同位素在自然界很少，^{13}C 约 1.1%，^{15}N 约 0.37%；丰度高的 ^{14}N 有四极矩，信号宽到没法用。办法是用细菌（通常是大肠杆菌）表达蛋白，培养基里唯一的氮源是 $^{15}NH_4Cl$，唯一的碳源是 ^{13}C 葡萄糖，这样产物里每个氮、每个碳都是自旋 ½ 的核。碳的信号提高约一百倍，氮提高近三百倍，更重要的是磁化可以借助一键耦合沿主链逐个原子传递，后面所有三共振实验都建立在这一点上。

for carbon and nearly three hundredfold for nitrogen, and magnetization can be passed from atom to atom along the backbone through one-bond couplings, which is the basis of every triple-resonance experiment that follows.

SLIDE 43 The $^{1}H–^{15}N$ HSQC: A Protein's Fingerprint $^{1}H–^{15}N$ HSQC：蛋白质的指纹谱



IN ONE LINE · 一句话

One peak per backbone N–H, so roughly one peak per residue.

每个主链 N–H 一个峰，大致每个残基一个峰。

$^{1}H–^{15}N$ HSQC 氢–氮 HSQC　**amide** 酰胺　**folded / aliased peak** 折叠峰　**fingerprint** 指纹

EN

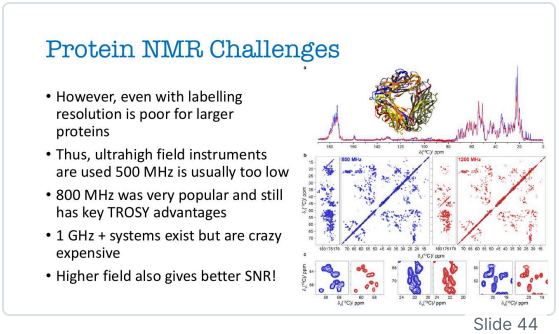
In small-molecule work HSQC meant $^{1}H–^{13}C$. For proteins the standard is $^{1}H–^{15}N$. Each residue has exactly one backbone amide N–H (highlighted in the diagram), so each residue gives one peak, and the ^{15}N dimension spreads those peaks over about 30 ppm where the proton dimension alone offers barely 3 ppm. The annotated spectrum is worth learning. The main cloud holds the backbone amides, with glycines toward the top at low ^{15}N shift. Top right, circled in purple: side-chain NH_2 groups of Asn and Gln, which appear as pairs of peaks sharing one nitrogen shift at two proton shifts. Lower left, in blue: tryptophan indole Hε–Nε, near 10 ppm in ^{1}H. Bottom, in green: arginine Hε–Nε peaks, which truly lie near 87 ppm in ^{15}N, outside the spectral window, and are therefore folded in with opposite sign. The C-terminal residue often gives an unusually strong peak. Residues with no peak: proline, which has no amide H, and the N-terminal residue.

中文

做小分子时 HSQC 指 $^{1}H–^{13}C$；做蛋白，标准配置是 $^{1}H–^{15}N$。每个残基正好有一个主链酰胺 N–H（示意图里标红的部分），所以每个残基出一个峰；氢维本身只有不到 3 ppm 的展开范围，^{15}N 维把这些峰摊开到约 30 ppm。那张带注释的谱图值得记住。中间的一大团是主链酰胺，甘氨酸偏上方（^{15}N 位移低）。右上紫圈：Asn 和 Gln 侧链的 NH_2，表现为氮位移相同、氢位移不同的成对峰。左下蓝圈：色氨酸吲哚环的 Hε–Nε，氢位移在 10 ppm 附近。下方绿圈：精氨酸的 Hε–Nε，真实的 ^{15}N 位移约 87 ppm，超出谱宽，被折叠进来，符号相反。C 端残基常给出一个特别强的峰。不出峰的残基：脯氨酸（没有酰胺氢）和 N 端残基。

要点 · Tip　HSQC 能直接读出的信息：峰数是否约等于残基数（样品是否均一）；峰是否分散（是否折叠）；加配体后哪些峰移动（结合位点在哪）。

SLIDE 44 Solution 2: Higher Field 对策二：更高的磁场



Slide 44

IN ONE LINE · 一句话

Higher field spreads peaks apart in Hz and raises sensitivity; proteins start at 600 MHz and go past 1 GHz.

磁场越高，峰在 Hz 上分得越开，灵敏度也越高；蛋白质 NMR 从 600 MHz 起步，直到 1 GHz 以上。

field strength 磁场强度 · resolution 分辨率 · SNR 信噪比 · ultrahigh field 超高场

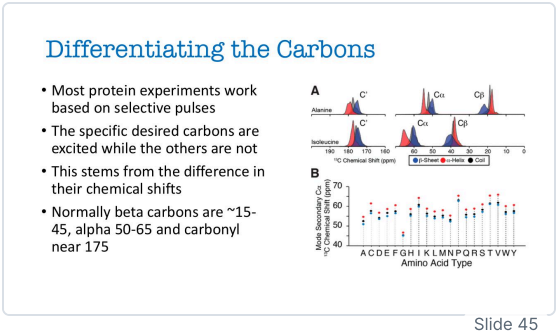
EN

Labelling alone does not resolve a large protein. Chemical shift differences in Hz scale with the field, while many contributions to linewidth do not, so peaks separate better at higher field. Sensitivity also rises, roughly as B_o to the power 3/2. That is why the slide calls 500 MHz usually too low, why 800 MHz became the workhorse (it also sits near the field where TROSY works well), and why instruments at 1 GHz and above exist despite their cost; 1.2 GHz is the current commercial maximum. The figure compares spectra of one large protein at 850 MHz (blue) and 1200 MHz (red): in the enlarged regions at the bottom, peaks that merge at 850 MHz are separate at 1200 MHz.

中文

对大蛋白来说，光靠标记还分不开。化学位移差换算成 Hz 与磁场成正比，而线宽的很多来源并不随场强增加，所以场越高峰分得越开。灵敏度也随之提高，大致正比于 B_o 的 3/2 次方。所以讲义说 500 MHz 通常太低，800 MHz 成了主力机型（它也接近 TROSY 效果好的场强），1 GHz 以上的仪器尽管贵得离谱还是有人造；目前商品化的最高场是 1.2 GHz。图中比较了同一个大蛋白在 850 MHz（蓝）和 1200 MHz（红）下的谱：底部的放大区域里，850 MHz 下粘在一起的峰在 1200 MHz 下分开了。

SLIDE 45 Telling the Carbons Apart 区分三类碳



Slide 45

IN ONE LINE · 一句话

C′ ≈ 175 ppm, Cα ≈ 50–65 ppm, Cβ ≈ 15–45 ppm: far enough apart to be excited selectively.

C′ 约 175 ppm，Cα 约 50–65 ppm，Cβ 约 15–45 ppm：相距足够远，可以分别选择性激发。

selective pulse 选择性脉冲 · carbonyl carbon (C′) 羰基碳 · Cα / Cβ α 碳 / β 碳 · secondary chemical shift 二级化学位移

EN

Triple-resonance experiments must send magnetization to one kind of carbon and not another. They can, because the three backbone carbon types occupy separate regions: carbonyl C′ near 175 ppm, Cα at 50 to 65 ppm, Cβ at about 15 to 45 ppm. A shaped, band-selective pulse excites one region and leaves the others untouched, so the spectrometer treats C′ and Cα almost as different nuclei. Panel A of the figure shows something more: the same residue type has slightly different shifts in a helix (red) and in a strand (blue). Cα and C′ move to higher ppm in helices and to lower ppm in strands, and Cβ does the

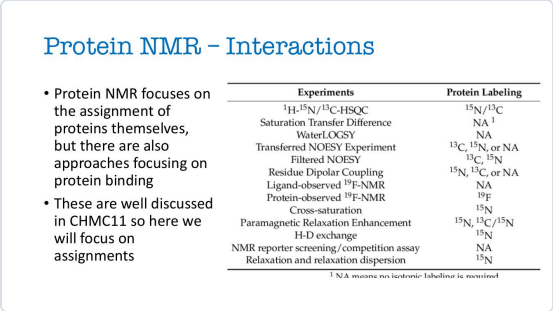
中文

三共振实验必须能把磁化传给某一类碳而不传给另一类。之所以做得到，是因为主链上三类碳各占一段区域：羰基 C′ 在 175 ppm 附近，Cα 在 50–65 ppm，Cβ 大约在 15–45 ppm。用整形的带选择性脉冲可以只激发其中一段，不碰其他，谱仪几乎是把 C′ 和 Cα 当成两种不同的核来处理。图 A 还透露了更多信息：同一种残基，在螺旋里（红）和在 β 链里（蓝）位移略有不同。Cα 和 C′ 在螺旋中移向高 ppm、在 β 链中移向低 ppm，Cβ 正好相反。主链归属完成之后，这些偏差不需要额外实验就能给出二级结构。图 B 是每种氨基酸典型的 Cα 位移，甘氨酸在 45 ppm 附近，明显偏离其他残基。

opposite. Once the backbone is assigned, these deviations give the secondary structure with no further experiments. Panel B shows the typical Cα shift of each amino acid; glycine, near 45 ppm, is the clear outlier.

> **要点 · Tip** 三个有用的例外，都能直接认出残基类型：甘氨酸没有 Cβ，Cα ≈ 45；丙氨酸 Cβ ≈ 19，是最低的；丝氨酸和苏氨酸的 Cβ 连着氧，分别在约 64 和约 70 ppm，比它们自己的 Cα 还高。

SLIDE 46 Protein NMR for Interactions 用 NMR 研究相互作用



IN ONE LINE · 一句话

A separate family of experiments studies binding; many need no labelling. Covered in CHMC11.

另有一大类实验研究结合，很多不需要标记。这部分在 CHMC11 讲。

saturation transfer difference (STD) 饱和转移差谱 · **WaterLOGSY** WaterLOGSY · **H–D exchange** 氢氘交换 · **ligand- / protein-observed** 观测配体 / 观测蛋白

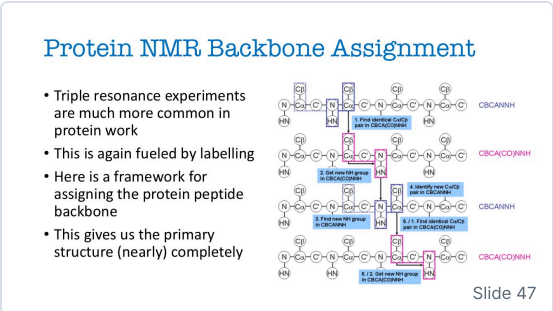
EN

This lecture is about assignment, but the table is a reminder that much protein NMR asks a different question: does this molecule bind, and where? The right-hand column shows what each experiment costs in labelling. Ligand-observed methods such as saturation transfer difference and WaterLOGSY watch the small molecule and need no labelled protein ("NA"), which makes them popular for drug screening. Protein-observed methods such as ^{1}H–^{15}N HSQC titrations need ^{15}N-labelled protein and reveal which residues sense the ligand. ^{19}F experiments exploit the fact that biology contains essentially no fluorine, so there is no background.

中文

这一讲讲的是归属，但这张表提醒我们，蛋白质 NMR 里有很大一部分在问另一个问题：这个分子结不结合，结合在哪里？表的右栏是每种实验对标记的要求。观测配体的方法，如饱和转移差谱和 WaterLOGSY，看的是小分子，不需要标记蛋白（表中写 “NA”），所以在药物筛选里很常用。观测蛋白的方法，如 ^{1}H–^{15}N HSQC 滴定，需要 ^{15}N 标记的蛋白，能看出哪些残基感受到了配体。^{19}F 实验利用的是生物体系里基本不含氟，因此没有背景信号。

SLIDE 47 Backbone Assignment: The Framework 主链归属：整体思路



IN ONE LINE · 一句话

Use each residue's N–H as an anchor and ask two questions: what are my carbons, and what are my predecessor's? 以每个残基的 N–H 为锚点，问两个问题：我自己的碳是什么？我前一个残基的碳是什么？

triple resonance 三共振 · **backbone assignment** 主链归属 · **sequential walk** 顺序行走 · **residue i / i–1** 第 i 个 / 第 i–1 个残基

EN

A triple-resonance experiment correlates three kinds of nuclei, ^{1}H, ^{15}N and ^{13}C, in one 3D spectrum, which is possible only with a doubly labelled protein. The assignment strategy pairs two experiments. For the N–H of residue i, CBCANNH shows the Cα and Cβ of residue i and those of residue i–1. CBCA(CO)NNH shows only those of residue i–1. The flow chart on the slide

中文

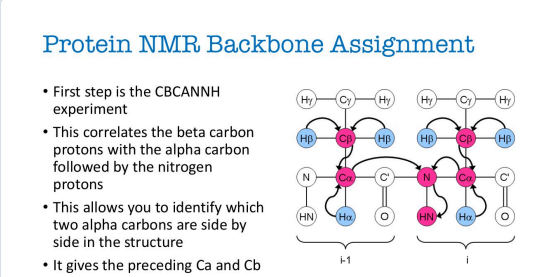
三共振实验在一张三维谱里把 ^{1}H、^{15}N、^{13}C 三种核关联起来，只有双标记的蛋白才能做。归属策略是两个实验配对使用。对第 i 个残基的 N–H：CBCANNH 同时给出第 i 个残基和第 i–1 个残基的 Cα、Cβ；CBCA(CO)NNH 只给出第 i–1 个残基的。讲义上的流程图就是在两张谱之间来回：在一张谱里找到一对 Cα/Cβ，再到另一张谱里

alternates between the two: find a Cα/Cβ pair in one spectrum, find the N–H that sees the same pair in the other, and step along the chain. Each step links one residue to its neighbour, and the Cα/Cβ shifts indicate the residue type, so a run of linked residues can be placed on the known sequence. The result is the primary structure with resonance assignments attached. The slide says "nearly" because prolines, which have no N–H, interrupt the walk, and overlap can leave gaps.

找看到同一对碳的 N–H，这样沿着链一步步走。每一步把一个残基和它的邻居连起来，而 Cα/Cβ 的位移又能提示残基类型，所以一串连起来的残基可以对到已知序列上。结果是带有共振归属的一级结构。讲义说「几乎」，是因为脯氨酸没有 N–H，会把行走打断，谱峰重叠也会留下缺口。

要点 · Tip 名字的读法：按磁化经过的原子顺序写；括号里的原子只「路过」，不记录它的化学位移。CBCA(CO)NNH = 从 Cβ/Cα 出发，经过 C′（不记录），到 N，再到 HN。

SLIDE 48 Experiment 1: CBCANNH 实验一：CBCANNH



IN ONE LINE · 一句话

Each N–H sees its own Cα/Cβ and, more weakly, the Cα/Cβ of the residue before it.

每个 N–H 看到自己的 Cα/Cβ，也较弱地看到前一个残基的 Cα/Cβ。

CBCANH / HNCACB 同一类实验的不同写法 · **intra-residue** 残基内 · **sequential (i–1)** 顺序相关 · **J coupling** J 耦合

EN

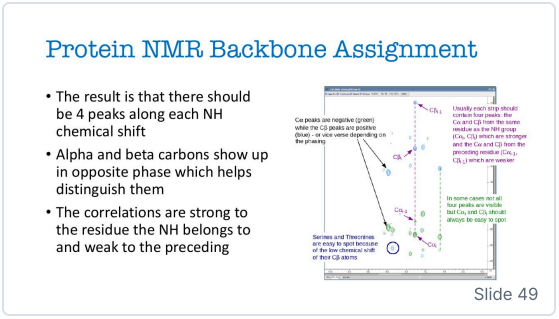
Follow the arrows in the diagram. Magnetization starts on the Hα and Hβ protons, moves to the carbons they are attached to, gathers on Cα, crosses to nitrogen and ends on the amide proton, where it is detected. The three axes of the spectrum are ^{13}C (Cα and Cβ), ^{15}N and ^{1}H(N). The essential feature is that Cα is coupled to two nitrogens: its own, through one bond (about 11 Hz), and the nitrogen of the next residue, through two bonds across the carbonyl (about 7 Hz). Magnetization from the Cα and Cβ of residue i–1 therefore also reaches the N–H of residue i. So each N–H reports two residues: its own and its predecessor. The same correlations are often recorded under the name HNCACB, which starts and ends on the amide proton; you will see both names.

中文

跟着示意图里的箭头走。磁化从 Hα、Hβ 出发，转到它们所连的碳上，汇集到 Cα，跨到氮，最后到酰胺氢上被检测。谱图的三个轴是 ^{13}C（Cα 和 Cβ）、^{15}N、^{1}H(N)。关键在于 Cα 同时与两个氮耦合：一键耦合到自己的氮（约 11 Hz），以及隔着羰基、两键耦合到下一个残基的氮（约 7 Hz）。所以第 i–1 个残基的 Cα、Cβ 上的磁化也能到达第 i 个残基的 N–H。于是每个 N–H 报告两个残基：自己和前一个。同样的相关也常用 HNCACB 这个名字来记录，它从酰胺氢出发又回到酰胺氢，两个名字都会遇到。

留意 · Check 讲义最后一条 “It gives the preceding Ca and Cb” 容易读成「只给出前一个」。完整的说法是：它既给出本残基的，也给出前一个残基的。只给前一个的是下一页的 CBCA(CO)NNH。

SLIDE 49 Reading CBCANNH: Four Peaks per Strip 读 CBCANNH：每条有四个峰



IN ONE LINE · 一句话

Two strong (own Cα, Cβ) + two weak (previous Cα, Cβ); Cα and Cβ have opposite sign.

两强（自己的 Cα、Cβ）加两弱（前一个的 Cα、Cβ）；Cα 与 Cβ 符号相反。

strip 条带 · phase / sign 相位 / 符号 · strong vs weak 强峰与弱峰

EN

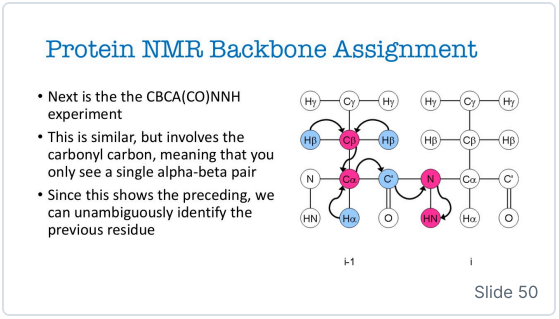
A 3D spectrum is read in strips: narrow slices taken at the ^{1}H and ^{15}N shifts of one HSQC peak, with ^{13}C running vertically. In a CBCANNH strip you expect four peaks, as annotated on the slide. Sign separates carbon type: Cα peaks have one sign and Cβ peaks the other (green and blue here; which is which depends on phasing). Intensity separates residue: the pair from the N–H's own residue is strong because it arrives through the larger one-bond coupling, and the pair from residue i–1 is weak. The annotations add practical notes. Sometimes not all four peaks are visible, but the residue's own Cα and Cβ should always be easy to find. Glycine gives no Cβ. Serine and threonine stand out because their Cβ lies at unusually high ppm, close to or above the Cα region.

中文

三维谱是按「条带」读的：在某个 HSQC 峰的 ^{1}H 和 ^{15}N 位移处切出窄窄的一条，纵轴是 ^{13}C。CBCANNH 的一条里应该有四个峰，讲义图上有注释。符号区分碳的类型：Cα 峰是一种符号，Cβ 峰是另一种（图中是绿和蓝，哪个对应哪个取决于调相）。强度区分残基：属于这个 N–H 自己残基的那一对强，因为走的是较大的一键耦合；来自第 i–1 个残基的那一对弱。图上的注释还有几条实用提示：有时四个峰看不全，但自己残基的 Cα、Cβ 应该总是容易找到；甘氨酸没有 Cβ；丝氨酸和苏氨酸很显眼，因为它们的 Cβ 位移特别高，接近甚至超过 Cα 区。

留意 · Check ① 图中注释说丝氨酸、苏氨酸因 Cβ 的 “low chemical shift” 而容易辨认。按数值看，它们的 Cβ 位移其实很高（约 64 和 70 ppm），属于低场（downfield）；这张谱的 ^{13}C 轴是向下增大的，所以它们出现在图的下方。这里的 “low” 应理解为低场或图上位置靠下，而不是 ppm 数值小。② 强弱只是经验规律，偶尔两对峰强度相近，这时无法单凭强度判断哪对属于前一个残基。下一个实验正是为了消除这种歧义。

SLIDE 50 Experiment 2: CBCA(CO)NNH 实验二：CBCA(CO)NNH



IN ONE LINE · 一句话

Routing through the carbonyl makes the transfer one-way: only the previous residue's Cα and Cβ appear.

让磁化经过羰基，传递就成了单行道：只出现前一个残基的 Cα 和 Cβ。

CBCA(CO)NH 常见写法 · carbonyl relay 经羰基中继 · unambiguous 无歧义

EN

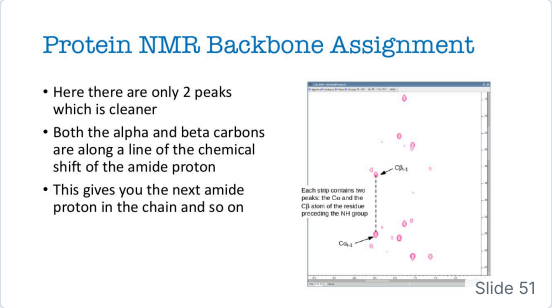
The start is the same, Hβ/Hα to Cβ/Cα to Cα, but the magnetization is then sent from Cα to the carbonyl carbon and from there to nitrogen. A carbonyl is bonded to the Cα of its own residue and to the nitrogen of the next residue. So magnetization that began on residue i–1 can end only on the N–H of residue i; there is no route from a residue's own carbons to

中文

开头一样，Hβ/Hα → Cβ/Cα → Cα，但接下来磁化从 Cα 传到羰基碳，再从羰基碳传到氮。羰基碳一边连着本残基的 Cα，另一边连着下一个残基的氮，所以从第 i–1 个残基出发的磁化只能落到第 i 个残基的 N–H 上；在这个实验里，残基自己的碳没有路径通向自己的 N–H。CO 写在括号里，是因为它的位移不被记录。结果是条带里只有前一个残基的 Cα 和 Cβ，不依赖峰的强弱就能确定前一个残基是谁。

its own N–H in this experiment. The CO is in parentheses because its shift is not recorded. The result is a strip containing only the Cα and Cβ of the preceding residue, which identifies the predecessor without relying on peak intensity.

SLIDE 51 Reading CBCA(CO)NNH: Two Peaks per Strip 读 CBCA(CO)NNH：每条两个峰



IN ONE LINE · 一句话

Two peaks = Cα(i−1) and Cβ(i−1). Subtract them from the four and the remaining two are your own.

两个峰 = Cα(i−1) 和 Cβ(i−1)。从四个峰里去掉它们，剩下的两个就是自己的。

sequential connectivity 顺序连接 · matching shifts 位移匹配

EN

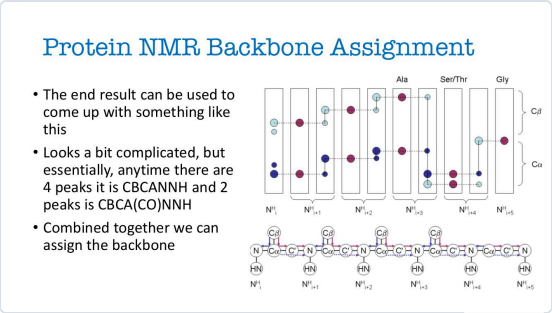
Each strip now has two peaks, Cα and Cβ of residue i−1, at the same $^{1}H/^{15}N$ position as the four-peak strip from CBCANNH. Overlay the two strips: the two peaks that appear in both belong to the previous residue, and the two that appear only in CBCANNH belong to the N–H's own residue. To extend the chain, take the i−1 shifts and search all the other strips for the N–H whose own Cα/Cβ match them. That N–H is residue i−1. Its strip gives the carbons of residue i−2, and the process repeats. Matching on two carbons at once makes accidental coincidences much less likely than matching on Cα alone.

中文

现在每条只有两个峰，就是第 i−1 个残基的 Cα 和 Cβ，位置与 CBCANNH 四峰条带的 $^{1}H/^{15}N$ 坐标相同。把两条叠起来看：两张谱里都有的两个峰属于前一个残基，只在 CBCANNH 里出现的两个峰属于这个 N–H 自己的残基。要把链接长，就拿着 i−1 的位移，到其他所有条带里去找「自己的 Cα/Cβ」与之相符的那个 N–H，它就是第 i−1 个残基。它的条带又给出第 i−2 个残基的碳，如此重复。同时匹配两个碳，比只匹配 Cα 更不容易碰巧撞上。

留意 · Check 讲义写 “gives you the next amide proton in the chain”。按实验给出的信息，行走方向是朝 i−1（朝 N 端）；倒过来问「谁的条带把我当作前一个残基」，就能朝 C 端走。两个方向都可以，关键是分清条带里哪对峰属于自己。

SLIDE 52 Putting It Together: The Strip Plot 合在一起：条带图



IN ONE LINE · 一句话

Line the strips up in sequence order; horizontal lines join each residue's own peaks to the next strip's i−1 peaks.

把条带按序列排好；水平线把每个残基自己的峰与下一条里的 i−1 峰连起来。

strip plot 条带图 · residue type 残基类型 · mapping to sequence 对应到序列

EN

This schematic is the finished product. Strips come in pairs for each N–H: the four-peak CBCANNH strip and the two-peak CBCA(CO)NNH strip, as the slide's rule of thumb says. Dotted horizontal lines connect a carbon shift in one residue's strip to the same shift in its neighbour's, tracing the walk along NH(i), NH(i+1), NH(i+2) and so on. The labels on top show how residue types are recognized from the pattern. Ala: the Cβ is far

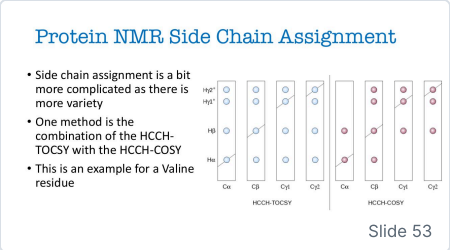
中文

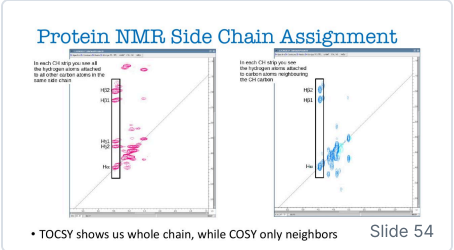
这张示意图就是最终成果。每个 N–H 都有成对的两条：四个峰的 CBCANNH 条带和两个峰的 CBCA(CO)NNH 条带，正如讲义的口诀所说。水平虚线把一个残基条带里的某个碳位移和相邻残基条带里相同的位移连起来，描出沿 NH(i)、NH(i+1)、NH(i+2) 的行走路线。顶部的标注说明残基类型如何从图案中认出来。丙氨酸：Cβ 远

from everything else at about 19 ppm. Ser/Thr: Cβ in or beyond the Cα region. Gly: one carbon only, a Cα near 45 ppm. A run such as "X–Ala–X–Ser–Gly" will usually match only one place in the protein's sequence, and that match anchors the whole stretch. Appendix C works through a numerical example.

离其他峰，在约 19 ppm。丝氨酸/苏氨酸：Cβ 落在 Cα 区域内甚至更高。甘氨酸：只有一个碳，Cα 在 45 ppm 附近。像「X–Ala–X–Ser–Gly」这样的一小段，在蛋白序列里通常只有一处对得上，这一处就把整段都锚定了。附录 C 有一个带数字的完整例子。

SLIDES 53–54 Side-Chain Assignment: HCCH-TOCSY and HCCH-COSY 侧链归属：HCCH-TOCSY 与 HCCH-COSY





IN ONE LINE · 一句话

TOCSY shows every proton in the side chain; COSY shows only protons on the neighbouring carbon.

TOCSY 给出侧链上所有的氢；COSY 只给出相邻碳上的氢。

side chain 侧链 · spin system 自旋体系 · HCCH-TOCSY HCCH-TOCSY · HCCH-COSY HCCH-COSY

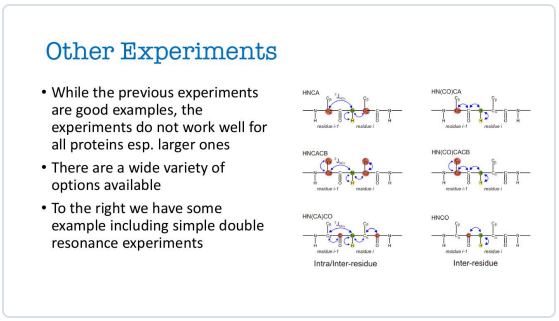
EN

After the backbone walk, each residue has assigned Cα and Cβ shifts. These are the entry points to the side chain, where the twenty amino acids differ and so the patterns vary more. The HCCH experiments pass magnetization from a proton to its carbon, along the carbon chain through the large one-bond C–C couplings (about 35 Hz), and back to a proton. In HCCH-TOCSY the carbon mixing spreads magnetization over the whole chain, so a strip taken at one C–H pair shows every proton of that side chain. In HCCH-COSY the transfer stops after one C–C bond, so the strip shows only the protons on the directly neighbouring carbon. The logic is the one used for small molecules: TOCSY tells you which signals belong together, COSY tells you their order. The schematic uses valine (Hα, Hβ, two γ-methyls): with TOCSY every strip contains all positions, with COSY each strip shows only its neighbours. The second slide shows real strips with the proton labels down the side.

中文

主链行走完成后，每个残基的 Cα、Cβ 位移都已归属，它们是进入侧链的入口。二十种氨基酸的差别就在侧链，所以图案变化更多。HCCH 类实验把磁化从氢传到所连的碳，沿碳链靠较大的一键 C–C 耦合（约 35 Hz）传递，再回到氢上。HCCH-TOCSY 里，碳上的混合让磁化传遍整条链，所以在某一个 C–H 对上取的条带能看到这条侧链上所有的氢。HCCH-COSY 里，传递只走一根 C–C 键，条带上只出现直接相邻碳上的氢。逻辑和做小分子时一样：TOCSY 告诉你哪些信号属于同一个体系，COSY 告诉你它们的先后顺序。示意图用的是缬氨酸（Hα、Hβ、两个 γ 甲基）：TOCSY 里每条都有所有位置的峰，COSY 里每条只有邻居。第二页是真实条带，旁边标着各个氢。

SLIDE 55 Other Triple-Resonance Experiments 其他三共振实验



Slide 55

IN ONE LINE · 一句话

A menu of experiments, in pairs: one sees residues i and i−1, its partner sees only i−1.

一组可选的实验，成对出现：一个同时看到 i 和 i−1，它的搭档只看到 i−1。

HNCA / HN(CO)CA HNCA / HN(CO)CA

HNCACB / HN(CO)CACB HNCACB / HN(CO)CACB

HN(CA)CO / HNCO HN(CA)CO / HNCO

intra-/inter-residue 残基内 / 残基间

EN

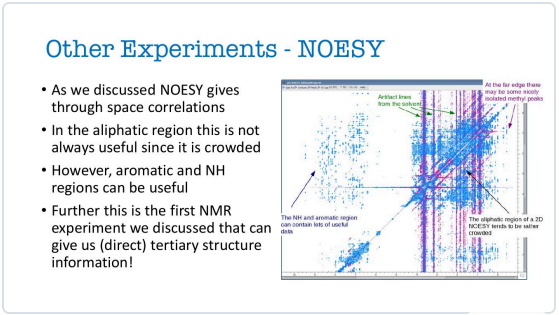
The Cα/Cβ pair of experiments is one option among several, and it becomes too insensitive for larger proteins. The figure lays out the standard set in two columns. Left column, "intra/inter-residue": HNCA, HNCACB and HN(CA)CO each show the carbon of residue i and of residue i−1. Right column, "inter-residue": HN(CO)CA, HN(CO)CACB and HNCO pass through the carbonyl and show only residue i−1. Every row is the same two-question strategy applied to a different carbon: Cα only, Cα plus Cβ, or C′. HNCO deserves a note: it starts on the amide proton, goes to N, then across the peptide bond to the C′ of residue i−1 and back. The slide also mentions double-resonance experiments, which involve only two kinds of nuclei, such as the ^{1}H–^{15}N HSQC or a ^{15}N-edited TOCSY.

中文

Cα/Cβ 这一对实验只是多种选择之一，蛋白大了以后它的灵敏度就不够了。图把常用的一套分成两栏。左栏「残基内/残基间」：HNCA、HNCACB、HN(CA)CO，各自同时给出第 i 个和第 i−1 个残基的碳。右栏「残基间」：HN(CO)CA、HN(CO)CACB、HNCO，都经过羰基，只给出第 i−1 个残基的。每一行都是同一套「两个问题」的策略，只是换了一种碳：只看 Cα、看 Cα 加 Cβ、或者看 C′。HNCO 值得单独说一句：它从酰胺氢出发到 N，跨过肽键到第 i−1 个残基的 C′，再原路返回。讲义还提到双共振实验，也就是只涉及两种核的实验，例如 ^{1}H–^{15}N HSQC 或 ^{15}N 编辑的 TOCSY。

要点 · Tip 附录 B 把这六个实验（外加讲义主讲的两个）整理成一张速查表：每个实验看到哪些碳、来自哪个残基、相对灵敏度如何。

SLIDE 56 NOESY: Through Space NOESY：空间相关



Slide 56

IN ONE LINE · 一句话

Cross peaks link protons closer than about 5 Å, however far apart they are in the sequence. This is where 3D structure comes from.

交叉峰连接空间距离约 5 Å 以内的氢，不管它们在序列上隔多远。三维结构就是从这里来的。

NOE 核奥弗豪泽效应

through-space 通过空间

distance restraint 距离约束

tertiary structure 三级结构

EN

Every experiment so far transfers magnetization through bonds, so it reveals connectivity and nothing about the fold. NOESY transfers magnetization through space by dipolar cross-relaxation. Its efficiency falls off as $1/r^6$, so a cross peak appears only when two protons are within about 5 Å, and a stronger peak means a shorter distance. Two residues fifty positions apart in the sequence that give an NOE must be neighbours in the folded protein. Hundreds or thousands of

中文

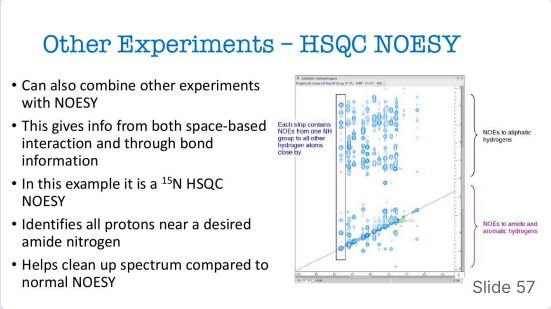
到目前为止，所有实验都是通过化学键传递磁化，只能给出连接关系，给不出折叠方式。NOESY 靠偶极交叉弛豫通过空间传递磁化，效率按 $1/r^6$ 衰减，所以只有当两个氢相距约 5 Å 以内时才有交叉峰，峰越强距离越近。序列上相隔五十个残基的两个残基如果有 NOE，在折叠好的蛋白里一定挨着。成百上千个这样的距离约束，

such distance restraints, combined with the known covalent geometry, are what a structure calculation uses to find the 3D fold. The annotated 2D spectrum shows the practical problem. The aliphatic region is hopelessly crowded and streaked with artefacts from the water signal. The amide and aromatic region is more useful, and the high-field corner sometimes has well-isolated methyl peaks.

加上已知的共价几何，就是结构计算用来求三维折叠的依据。带注释的二维谱展示了实际困难：脂肪区挤得没法看，还有水峰带来的条纹状伪峰；酰胺和芳香区更有用；高场一角有时会有孤立得很好的甲基峰。

要点 · Tip 考点：通过键的实验（COSY、TOCSY、HSQC、三共振）→ 归属和一级结构；通过空间的实验（NOESY、ROESY）→ 三级结构。讲义特别强调 NOESY 是课上第一个能直接给出三级结构信息的 NMR 实验。

SLIDE 57 ^{15}N HSQC-NOESY: Adding a Dimension ^{15}N HSQC-NOESY：加一个维度



IN ONE LINE · 一句话

Spread the NOESY out by the ^{15}N shift of each amide: one clean strip of NOEs per residue.

按每个酰胺的 ^{15}N 位移把 NOESY 摊开：每个残基一条干净的 NOE 条带。

3D NOESY-HSQC 三维 NOESY-HSQC　**^{15}N-edited** ^{15}N 编辑　**overlap** 重叠

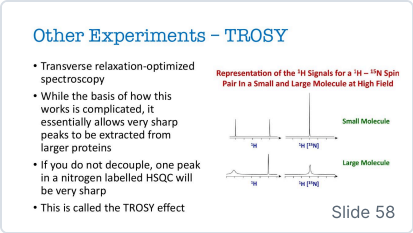
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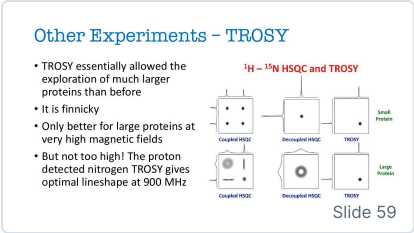
The cure for crowding is the same as before: add a dimension. A 3D ^{15}N NOESY-HSQC joins a NOESY step to a ^{1}H–^{15}N HSQC step, so every NOE to an amide proton is labelled with the shift of that amide's nitrogen. The spectrum is read in strips, one per HSQC peak. Each strip lists all the protons close in space to that one N–H: NOEs to aliphatic protons in the upper part and to other amide and aromatic protons in the lower part, with the diagonal peak of the amide itself as the reference. Because the residue that owns each N–H is already known from the backbone assignment, every cross peak in the strip is immediately a distance restraint starting from a known atom. The spectrum combines through-bond information (the HSQC step) with through-space information (the NOE step).

中文

对付拥挤的办法和之前一样：加一个维度。三维 ^{15}N NOESY-HSQC 把一步 NOESY 和一步 ^{1}H–^{15}N HSQC 接在一起，于是每个到酰胺氢的 NOE 都被标上了该酰胺氮的位移。谱图同样按条带读，每个 HSQC 峰一条。每条列出空间上靠近这个 N–H 的所有氢：上半部分是到脂肪氢的 NOE，下半部分是到其他酰胺氢和芳香氢的 NOE，酰胺氢自己的对角峰作为参照。因为每个 N–H 属于哪个残基已经由主链归属确定，条带里的每个交叉峰马上就是一条从已知原子出发的距离约束。这张谱把通过键的信息（HSQC 那一步）和通过空间的信息（NOE 那一步）结合了起来。

SLIDES 58–59 TROSY TROSY：横向弛豫优化谱





IN ONE LINE · 一句话

Skip decoupling, keep only the one multiplet component whose relaxation mechanisms cancel: a sharp peak even for a big protein.

不去耦，只保留多重峰里弛豫机制相互抵消的那一个分量：大蛋白也能出尖峰。

TROSY 横向弛豫优化谱 · **decoupling** 去耦 · **dipole–dipole relaxation** 偶极–偶极弛豫 · **chemical shift anisotropy (CSA)** 化学位移各向异性

EN

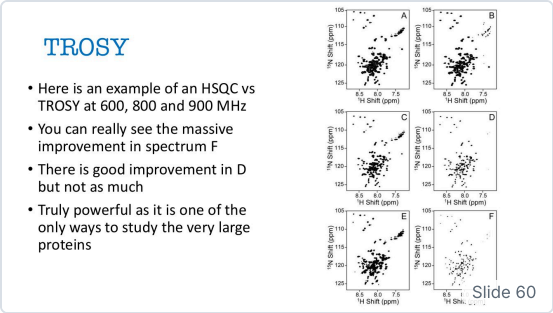
TROSY stands for transverse relaxation-optimized spectroscopy. In an ordinary HSQC the ^{1}H–^{15}N coupling is removed by decoupling, which collapses each peak's four multiplet components into one at the centre. Without decoupling you see all four (the "coupled HSQC" panels). For a small protein the four look alike. For a large protein they do not: one is sharp, one is very broad and two are in between. The reason is that an amide spin relaxes through two mechanisms, the dipole–dipole interaction with its bonded partner and its own chemical shift anisotropy. For one of the four components the two contributions have opposite sign and largely cancel; for the opposite component they add. Decoupling averages the fast- and slow-relaxing components and gives a broad peak. TROSY selects only the slowly relaxing component. Three quarters of the signal is discarded, yet for a large protein the surviving peak is so much narrower that both resolution and peak height improve. The second slide gives the conditions. The CSA contribution grows with field while the dipolar one does not, so cancellation is best at one particular field: about 900 MHz to 1 GHz for ^{1}H–^{15}N. Below that it is incomplete, and above it CSA overshoots. For a small protein the lines are already narrow, and TROSY merely costs signal.

中文

TROSY 是 transverse relaxation-optimized spectroscopy 的缩写。普通 HSQC 用去耦去掉 ^{1}H–^{15}N 耦合，把每个峰的四个多重峰分量合并成中间的一个。不去耦就能看到全部四个（图中“coupled HSQC” 一栏）。小蛋白的四个分量看起来差不多；大蛋白则不同：一个尖，一个很宽，另外两个介于两者之间。原因是酰胺自旋有两种弛豫机制：与成键原子之间的偶极–偶极相互作用，以及自身的化学位移各向异性。对四个分量中的一个，两种贡献符号相反，大部分抵消；对与之相对的那个分量，两者相加。去耦把弛豫快的和弛豫慢的分量平均到一起，得到宽峰。TROSY 只挑出弛豫慢的那个分量。虽然丢掉了四分之三的信号，但对大蛋白来说留下的峰窄得多，分辨率和峰高反而都更好。第二页给出了适用条件：CSA 的贡献随场强增大，偶极的贡献不随场强变化，所以两者在某个特定场强抵消得最好，对 ^{1}H–^{15}N 大约在 900 MHz 到 1 GHz；低于这个场强抵消不完全，高于它 CSA 又「过头」了。小蛋白的线本来就窄，用 TROSY 只会白白损失信号。

> **顺便一提** 讲义的评语 “It is finnicky” 很到位：TROSY 要求大蛋白、合适的场强，通常还要氘代，三者缺一不可。

SLIDE 60 TROSY in Practice: 600, 800, 900 MHz TROSY 实例：600、800、900 MHz



IN ONE LINE · 一句话

Left column HSQC, right column TROSY. The gain grows with field and is largest at 900 MHz (panel F).

左列 HSQC，右列 TROSY。场越高改善越明显，900 MHz（图 F）最显著。

lineshape 峰形 **field dependence 场强依赖**

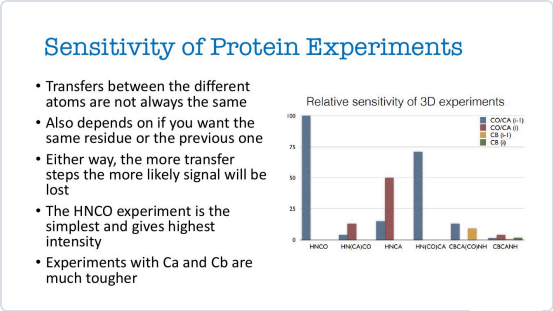
EN

The six panels show one large protein: conventional HSQC on the left (A, C, E) and TROSY on the right (B, D, F), at 600, 800 and 900 MHz from top to bottom. In the HSQC column the centre of the spectrum is a merged blob at every field. At 600 MHz, TROSY (B) changes little, because the two relaxation mechanisms are poorly matched there. At 800 MHz (D) individual peaks begin to separate. At 900 MHz (F) the blob resolves into many sharp, countable peaks. The trend matches the previous slide: TROSY pays off only when a large molecule is combined with a field close to the optimum. Together with deuteration of the non-exchangeable hydrogens, this is one of the few routes to solution NMR of very large proteins and complexes.

中文

六张小图是同一个大蛋白：左列（A、C、E）是常规 HSQC，右列（B、D、F）是 TROSY，从上到下依次是 600、800、900 MHz。HSQC 一列在每个场强下中间都是糊成一团。600 MHz 时 TROSY（B）变化不大，因为这个场强下两种弛豫机制匹配得不好；800 MHz（D）时单个峰开始分开；900 MHz（F）时那一团分解成许多尖锐、可以数清的峰。这个趋势和上一页一致：只有「大分子」加上「接近最佳值的场强」，TROSY 才划算。再配合对不可交换氢的氘代，这是溶液 NMR 研究超大蛋白和复合物的少数途径之一。

SLIDE 61 Sensitivity of 3D Experiments 三维实验的灵敏度



IN ONE LINE · 一句话

Every transfer step loses signal: HNCO (100) ≫ HN(CO)CA (~70) > HNCA (~50) ≫ the Cβ experiments.

每多一步传递就多一份损失：HNCO（100）远大于 HN(CO)CA（约 70）大于 HNCA（约 50），Cβ 实验最差。

magnetization transfer 磁化传递 **transfer efficiency 传递效率** **relative sensitivity 相对灵敏度**

EN

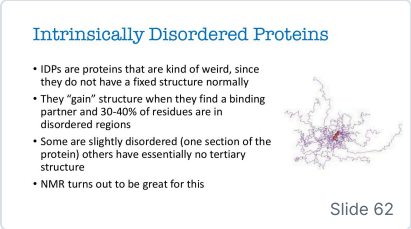
The bar chart ranks the standard experiments with HNCO set to 100. Two factors decide the ranking. The first is coupling size: a transfer through a large coupling is fast, and a fast transfer leaves less time for relaxation. N to C′ is about 15 Hz and Cα to C′ about 55 Hz, both favourable. N to Cα is only about 11 Hz, slow and lossy. The second is the number of steps: every extra transfer and every extra delay costs signal. HNCO has the shortest and most efficient path, H to N to C′ and back. HN(CO)CA, at about 70, adds one efficient step. HNCA, at about 50 for the intra-residue peak and about 15 for the sequential one, depends on the weak N–Cα couplings. Experiments that reach Cβ, namely CBCA(CO)NH and CBCANH, are at the bottom, around 10 or below, because the

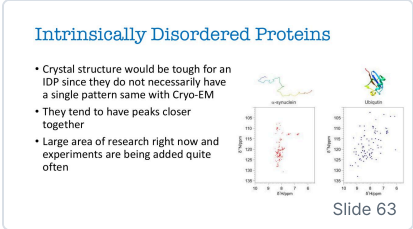
中文

柱状图以 HNCO 为 100，给常用实验排了名。排名由两个因素决定。一是耦合常数的大小：耦合大，传递快，留给弛豫的时间就少。N–C′ 约 15 Hz，Cα–C′ 约 55 Hz，都有利；N–Cα 只有约 11 Hz，又慢损失又大。二是步数：每多一步传递、多一段延迟，都要损失信号。HNCO 路径最短最高效：H → N → C′ 再返回。HN(CO)CA 约 70，只多了一步高效传递。HNCA 残基内的峰约 50、顺序峰约 15，因为它依赖较弱的 N–Cα 耦合。要到达 Cβ 的实验，CBCA(CO)NH 和 CBCANH，排在最后，只有 10 左右甚至更低，因为磁化还要在 Cα 和 Cβ 之间分配。这一讲用来做归属的正是这两个实验：每张谱的信息量最大，信号却最弱，所以蛋白一大就不得不改用 HNCA、HNCO 这一类。

magnetization is also divided between Cα and Cβ. These are the experiments this lecture used for assignment; they carry the most information per spectrum and have the least signal, which is why larger proteins force a move to the HNCA and HNCO family.

SLIDES 62–63 Intrinsically Disordered Proteins 固有无序蛋白





IN ONE LINE · 一句话

No fixed fold, so no crystal and no single cryo-EM shape; NMR handles them well.

没有固定折叠，结不出晶体，冷冻电镜也看不到单一形状；NMR 却很适合。

- **intrinsically disordered protein (IDP)** 固有无序蛋白
- **conformational ensemble** 构象系综
- **folding upon binding** 结合诱导折叠
- **α-synuclein** α-突触核蛋白

EN

An intrinsically disordered protein has no single stable tertiary structure under normal conditions; it exists as a rapidly interconverting ensemble of conformations, like the overlaid chains in the figure. Disorder is common: the slide notes that 30 to 40% of residues lie in disordered regions. Some proteins have one floppy segment, others have essentially no tertiary structure at all. Many fold only when they meet a binding partner. Crystallography and cryo-EM both depend on many copies having the same shape, so both fail here. Solution NMR does not need a fixed shape, and the fast internal motion of a disordered chain gives sharp lines even for a long sequence. The difficulty is dispersion. The two HSQC spectra on the second slide show it: folded ubiquitin spreads over roughly 6.5 to 9.5 ppm in ^{1}H, while disordered α-synuclein is squeezed into a band less than 1 ppm wide near 8 to 8.5 ppm, because every residue experiences the same averaged, solvent-exposed environment. The ^{15}N dimension still separates the peaks, and newer experiments, including ones that detect ^{13}C directly, are being developed for exactly this kind of protein.

中文

固有无序蛋白在正常条件下没有单一稳定的三级结构，而是以快速相互转换的构象系综存在，就像图里叠在一起的那些链。无序很普遍：讲义指出 30–40% 的残基处在无序区域。有的蛋白只有一段是松散的，有的基本没有三级结构。很多无序蛋白只在遇到结合对象时才折叠。晶体学和冷冻电镜都依赖大量拷贝具有相同形状，所以在这里都失效。溶液 NMR 不要求形状固定，而且无序链内部运动快，即使序列很长线也很尖。难点在分散度。第二页的两张 HSQC 说明了这一点：折叠的泛素在氢维上分布于大约 6.5–9.5 ppm，无序的 α-突触核蛋白则挤在 8–8.5 ppm 附近不到 1 ppm 宽的窄带里，因为每个残基感受到的都是同样的、平均化的、暴露在溶剂中的环境。^{15}N 维仍然能把峰分开，针对这类蛋白的新实验（包括直接检测 ^{13}C 的实验）也在不断出现。

> **要点 · Tip** 看到一张氢维只占 8–8.5 ppm 的 ^{1}H–^{15}N HSQC，第一反应应当是：这个蛋白无序或者没折叠好。

PART 6

Nucleic Acids

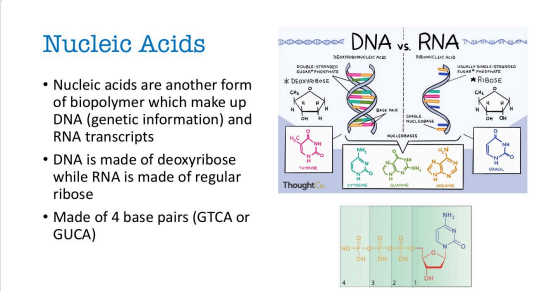
核酸

Slides 64–69

Sequence comes from sequencing, not spectroscopy; XRD and NMR supply the 3D picture.

序列靠测序而不是波谱；三维结构由 XRD 和 NMR 提供。

SLIDE 64 Nucleic Acids: DNA and RNA 核酸：DNA 与 RNA



Slide 64

IN ONE LINE · 一句话

Polymers of nucleotides: phosphate + sugar + base. DNA has deoxyribose and T; RNA has ribose and U.

核苷酸的聚合物：磷酸 + 糖 + 碱基。DNA 用脱氧核糖和 T；RNA 用核糖和 U。

nucleotide 核苷酸 · **deoxyribose / ribose** 脱氧核糖 / 核糖 · **base pair** 碱基对 · **phosphodiester** 磷酸二酯键

EN

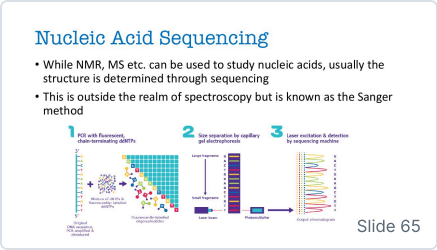
Nucleic acids are the second class of biopolymer. Each monomer, a nucleotide, has three parts: a phosphate, a five-carbon sugar and a nitrogenous base (the lower figure numbers the parts of a nucleoside triphosphate). Phosphodiester bonds between the 3′ carbon of one sugar and the 5′ carbon of the next form the backbone. DNA stores genetic information and uses 2′-deoxyribose and the bases G, C, A and T. RNA carries transcripts and has many other roles; it uses ribose, which keeps the 2′-OH, and has U in place of T. DNA is usually a double helix of two complementary strands. RNA is usually single-stranded and folds back on itself into hairpins and more elaborate shapes.

中文

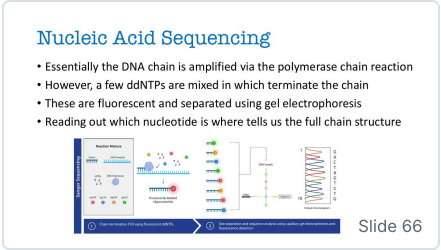
核酸是第二类生物高分子。每个单体（核苷酸）由三部分组成：磷酸、五碳糖、含氮碱基（下方的图给一个核苷三磷酸的各部分编了号）。一个糖的 3′ 碳和下一个糖的 5′ 碳之间的磷酸二酯键构成主链。DNA 储存遗传信息，用 2′-脱氧核糖，碱基是 G、C、A、T。RNA 是转录产物，还有许多其他功能，用核糖（保留 2′-OH），以 U 代替 T。DNA 通常是两条互补链形成的双螺旋；RNA 通常是单链，自身回折成发夹和更复杂的形状。

留意 · Check 讲义写 “Made of 4 base pairs (GTCA or GUCA)”。准确地说是 4 种碱基（bases）；碱基对（base pairs）指配对关系，只有两类：A 配 T（RNA 中为 U），G 配 C。

SLIDES 65–66 Sequencing: The Sanger Method 测序：桑格法



Slide 65



Slide 66

IN ONE LINE · 一句话

Copy the DNA with a few chain-terminating, colour-coded ddNTPs; sort the fragments by length; read the colours in order.

复制 DNA 时掺入少量带颜色、会终止链延伸的 ddNTP；按长度分离片段；依次读颜色。

- **Sanger sequencing** 桑格测序
- **ddNTP (dideoxynucleotide)** 双脱氧核苷酸
- **chain termination** 链终止
- **capillary gel electrophoresis** 毛细管凝胶电泳
- **PCR** 聚合酶链式反应

EN

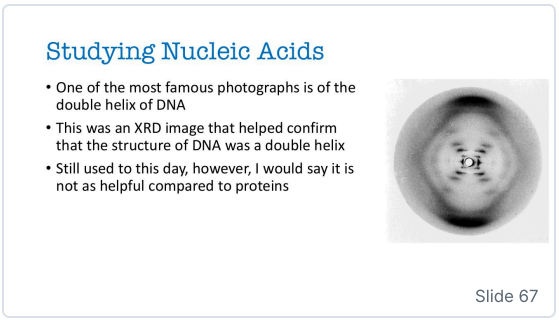
NMR and MS can study nucleic acids, but the primary structure is normally read by sequencing, which is not spectroscopy. The Sanger method works in three steps, as the figures show. Step 1: DNA polymerase copies the template in a reaction that contains the four normal dNTPs and a small amount of dideoxy versions, ddNTPs. A ddNTP has no 3′-OH, so once one is incorporated no further nucleotide can be attached and that copy stops. Because incorporation is random, the reaction yields copies terminated at every possible position. Each of the four ddNTPs carries a different fluorescent dye, so the colour of a fragment identifies its last base. Step 2: capillary gel electrophoresis separates the fragments by length with single-nucleotide resolution, shortest first. Step 3: a laser excites the dye as each fragment passes the detector. The sequence of colours in the output chromatogram is the sequence of bases.

中文

NMR 和 MS 都能研究核酸，但一级结构通常靠测序读出，而测序不属于波谱。桑格法分三步，对应讲义上的图。第一步：DNA 聚合酶复制模板，反应体系里有四种正常的 dNTP，另外掺入少量双脱氧的 ddNTP。ddNTP 没有 3′-OH，一旦掺入，后面就接不上新的核苷酸，这条拷贝到此为止。掺入是随机的，所以反应得到在每一个可能位置终止的拷贝。四种 ddNTP 各带一种不同的荧光染料，片段的颜色就说明它最后一个碱基是什么。第二步：毛细管凝胶电泳按长度分离片段，分辨率达到单个核苷酸，短的先出来。第三步：每个片段经过检测器时，激光激发染料。输出的色谱图里颜色的先后顺序，就是碱基的顺序。

> **要点 · Tip** 关键化学点：3′-OH 是形成下一个磷酸二酯键所必需的。ddNTP 缺少它，这就是「链终止」的来源。讲义把扩增步骤称为 PCR；测序反应本身是带终止剂的、类似 PCR 的热循环延伸反应。

SLIDE 67 XRD of DNA: Photo 51 DNA 的 X 射线衍射：照片 51 号



IN ONE LINE · 一句话

A fibre diffraction image; the X-shaped pattern is the signature of a helix.

一张纤维衍射图；X 形图案是螺旋的特征。

fibre diffraction 纤维衍射 · **double helix** 双螺旋 · **helical pitch** 螺距

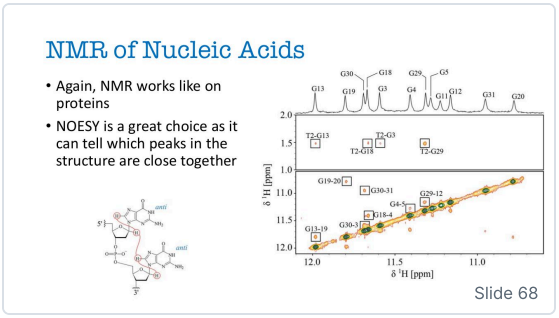
EN

The image is Photo 51, recorded in 1952 by Rosalind Franklin and Raymond Gosling from a hydrated DNA fibre. It is not a single-crystal pattern; the molecules in a fibre are aligned along one axis and otherwise disordered. Even so, the pattern can be read. The X-shaped cross of spots is what a helix produces. The spacing between the layers of spots gives the repeat distance along the helix (about 34 Å for one turn), and the strong dark arcs at top and bottom correspond to the 3.4 Å stacking distance between base pairs, so there are ten base pairs per turn. Watson and Crick used this evidence when they built their model. The slide adds that XRD is still used for nucleic acids but is less central than it is for proteins. One reason is that the Watson–Crick duplex looks much the same whatever its sequence; the informative cases are folded RNAs and complexes with proteins.

中文

这张图就是「照片 51 号」，1952 年由罗莎琳德·富兰克林和雷蒙德·戈斯林用含水的 DNA 纤维拍摄。它不是单晶衍射图：纤维里的分子沿一个轴取向排列，其他方向是无序的。即便如此，图案仍然可以解读。X 形交叉的斑点是螺旋会产生的图案；斑点层与层之间的间距给出沿螺旋轴的重复距离（一圈约 34 Å）；上下两端又黑又强的弧对应碱基对之间 3.4 Å 的堆积距离，所以每圈十个碱基对。沃森和克里克搭建模型时用到了这份证据。讲义补充说 XRD 至今仍用于核酸，但不像在蛋白质领域那样核心。原因之一是沃森–克里克双链不论序列如何，样子都差不多；真正有信息量的是折叠的 RNA 以及核酸与蛋白的复合物。

SLIDE 68 NMR of Nucleic Acids: NOESY 核酸 NMR：NOESY



IN ONE LINE · 一句话

Same toolkit as for proteins; NOEs between neighbouring bases give a sequential walk.

工具和蛋白质一样；相邻碱基之间的 NOE 可以用来做顺序行走。

imino proton 亚氨基氢 · **sequential NOE** 顺序 NOE · **anti conformation** 反式构象

EN

NMR of nucleic acids follows the same logic as protein NMR: identify spin systems with through-bond experiments, then connect them and determine the fold with NOESY. The spectrum on the slide shows the imino region, 10.8 to 12 ppm. Imino protons are the N–H protons of guanine and of thymine or uracil. They are visible only when they are protected from exchange with water, which happens when the base is hydrogen-bonded in a pair, so the number of imino peaks counts the base pairs that have formed. Stacked bases sit about 3.4 Å apart, inside the NOE range. The cross peaks labelled G19–20, G30–31, G29–12 and so on connect imino protons of neighbouring guanines, and each connection is one

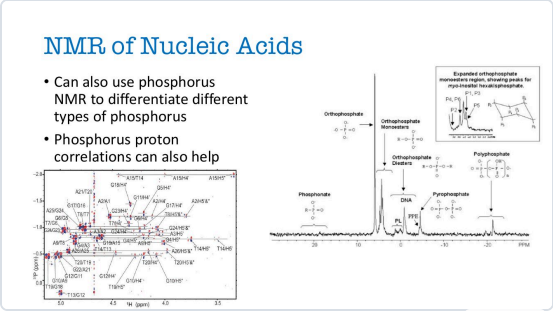
中文

核酸 NMR 的思路和蛋白质 NMR 相同：先用通过键的实验确定自旋体系，再用 NOESY 把它们连起来并确定折叠。讲义上的谱是亚氨基区，10.8–12 ppm。亚氨基氢是鸟嘌呤以及胸腺嘧啶或尿嘧啶上的 N–H。它们只有在不与水快速交换时才看得见，也就是碱基在配对中形成氢键的时候，所以亚氨基峰的数目反映了形成的碱基对数。堆积的碱基相距约 3.4 Å，在 NOE 的范围之内。标着 G19–20、G30–31、G29–12 等的交叉峰连接相邻鸟嘌呤的亚氨基氢，每一个

step along the stack. The upper panel shows NOEs from a thymine methyl (T2, near 1.5 ppm) to nearby guanine imino protons, which places T2 against those bases. The small structure at the lower left marks short proton–proton distances between two consecutive nucleotides in the anti conformation.

连接就是沿堆积方向走一步。上方的小图是某个胸腺嘧啶甲基（T2，约 1.5 ppm）到附近鸟嘌呤亚氨基氢的 NOE，说明 T2 贴着这些碱基。左下角的小结构图标出了两个相邻核苷酸（反式构象）之间距离较短的几对氢。

SLIDE 69 NMR of Nucleic Acids: ^{31}P 核酸 NMR：^{31}P



Slide 69

IN ONE LINE · 一句话

The backbone has one phosphorus per nucleotide; ^{31}P is 100% abundant and spin-½, a free extra nucleus.

主链上每个核苷酸有一个磷；^{31}P 丰度 100%、自旋 ½，是不用标记就能用的核。

- **^{31}P NMR** 磷谱
- **orthophosphate / pyrophosphate** 正磷酸 / 焦磷酸
- **phosphate monoester / diester** 磷酸单酯 / 二酯
- **^{1}H–^{31}P correlation** 氢–磷相关

EN

Nucleic acids have something proteins lack: a phosphorus atom in every backbone link. ^{31}P is spin-½ and 100% naturally abundant, so no labelling is needed, and its chemical shift responds to the kind of phosphorus group. The 1D spectrum on the right shows the range: phosphonates near +20 ppm, orthophosphate around +6, phosphate monoesters just upfield of it, diesters (where DNA appears) near 0, pyrophosphate around −5 and polyphosphate near −20. One spectrum therefore sorts the phosphorus in a sample by chemical type. The lower-left panel is a ^{1}H–^{31}P correlation of a DNA oligomer. Each phosphorus is coupled through three bonds to the H3′ of the sugar on one side and to the H5′/H5″ of the sugar on the other, so each ^{31}P signal links two consecutive nucleotides. That gives a through-bond sequential walk along the backbone, which complements the through-space NOESY walk on the previous slide.

中文

核酸有一样蛋白质没有的东西：主链的每个连接处都有一个磷原子。^{31}P 自旋 ½，天然丰度 100%，不需要标记，而且它的化学位移对磷所处基团的类型很敏感。右边的一维谱展示了大致范围：膦酸酯在 +20 ppm 附近，正磷酸约 +6，磷酸单酯在它的高场一侧不远处，二酯（DNA 出现的位置）在 0 附近，焦磷酸约 −5，多聚磷酸约 −20。所以一张谱就能把样品里的磷按化学类型分开。左下的小图是一段 DNA 寡聚体的 ^{1}H–^{31}P 相关谱。每个磷通过三键与一侧糖的 H3′ 耦合，又与另一侧糖的 H5′/H5″ 耦合，因此每个 ^{31}P 信号把两个相邻的核苷酸连在一起。这就得到了一条沿主链的、通过键的顺序行走路线，与上一页通过空间的 NOESY 行走互为补充。

PART 7

Complex Carbohydrates

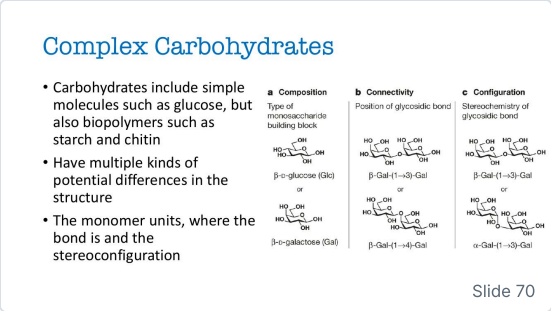
复杂糖类

Slides 70–74

The class with the most ways to vary: which sugar, which position, which configuration, and whether it branches.

变化方式最多的一类：哪种糖、连在哪个位置、什么构型、有没有分支。

SLIDE 70 Three Kinds of Variation 三个层面的差异



IN ONE LINE · 一句话

Composition (which monosaccharide), connectivity (which carbons are linked), configuration (α or β).

组成（哪种单糖）、连接方式（哪两个碳相连）、构型（α 还是 β）。

monosaccharide 单糖 · **glycosidic bond** 糖苷键 · **anomeric carbon** 异头碳 · **α / β configuration** α / β 构型 · **epimer** 差向异构体

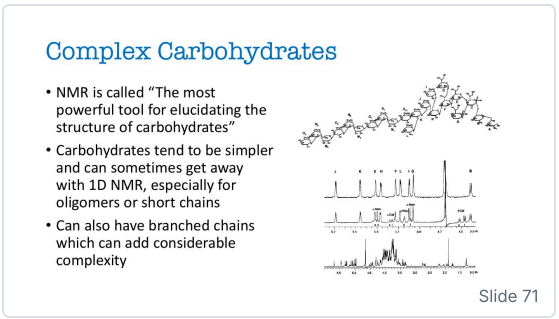
EN

Carbohydrates run from simple sugars such as glucose to polymers such as starch and chitin. The figure shows why their structures are harder to pin down than a protein's primary structure. (a) Composition: which monosaccharide? Glucose and galactose differ only in the orientation of the OH at C4, so they have the same mass and are indistinguishable by simple MS. (b) Connectivity: which hydroxyl of the second sugar carries the bond? Gal(1→3)Gal and Gal(1→4)Gal are different compounds. (c) Configuration: the anomeric carbon can be α or β. Starch is α(1→4)-linked glucose and cellulose is β(1→4)-linked glucose: the same monomer and the same position, with entirely different properties. A peptide bond always joins the same two functional groups in the same way, so proteins have none of this ambiguity. Two different amino acids can form two dipeptides; two hexoses can form dozens of disaccharides.

中文

糖类的范围从葡萄糖这样的单糖一直到淀粉、几丁质这样的聚合物。图说明了为什么糖的结构比蛋白质的一级结构更难确定。(a) 组成：是哪种单糖？葡萄糖和半乳糖只差 C4 上羟基的朝向，质量相同，简单的质谱分不开。(b) 连接方式：键连在第二个糖的哪个羟基上？Gal(1→3)Gal 和 Gal(1→4)Gal 是两种不同的化合物。(c) 构型：异头碳可以是 α 也可以是 β。淀粉是 α(1→4) 连接的葡萄糖，纤维素是 β(1→4) 连接的葡萄糖：单体相同、位置相同，性质完全不同。肽键总是以同一种方式连接同样的两个官能团，所以蛋白质没有这类歧义。两种不同的氨基酸只能形成两种二肽；两个己糖能形成几十种二糖。

SLIDE 71 Why NMR Suits Carbohydrates 为什么 NMR 适合糖类



IN ONE LINE · 一句话

NMR answers all three questions at once; short chains can sometimes be solved from 1D spectra.

NMR 能同时回答这三个问题；短链有时靠一维谱就能解决。

anomeric proton 异头氢 · oligosaccharide 寡糖 · branching 分支 · coupling constant ^3J 耦合常数 ^3J

EN

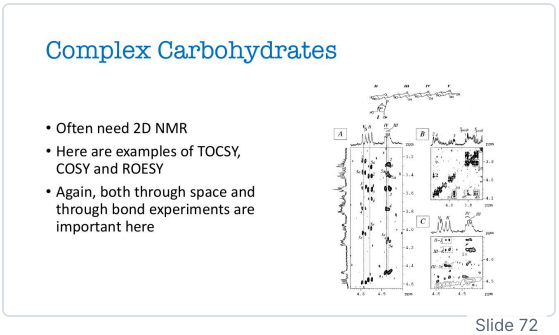
The slide quotes the view that NMR is the most powerful tool for carbohydrate structure, and the reason is that NMR is sensitive to stereochemistry. The starting point is the anomeric proton, H1 of each sugar ring. It sits on a carbon bonded to two oxygens, so it resonates at about 4.4 to 5.5 ppm, clear of the other ring protons, which crowd into 3.2 to 4.2 ppm. Counting anomeric signals gives the number of distinct sugar residues. The H1–H2 coupling gives the configuration: in glucose-type rings a β-anomer has H1 and H2 both axial and a coupling of 7 to 8 Hz, while an α-anomer has a coupling of 3 to 4 Hz. For oligomers and regular polymers the spectra are simple enough that 1D can be sufficient, as the slide says. Branching makes things harder: a branched residue carries substituents at two or more positions, and every branch adds a new set of overlapping ring signals, as the large structure in the figure suggests.

中文

讲义引用了「NMR 是解析糖类结构最有力的工具」这一说法，原因在于 NMR 对立体化学敏感。切入点是异头氢，也就是每个糖环上的 H1。它所在的碳连着两个氧，所以共振在约 4.4–5.5 ppm，与挤在 3.2–4.2 ppm 的其他环上氢分开。数一数异头氢信号，就知道有几种不同的糖残基。H1–H2 的耦合常数给出构型：在葡萄糖型的环里，β 异头体的 H1、H2 都处于直立位置，耦合为 7–8 Hz；α 异头体的耦合为 3–4 Hz。对寡糖和规整的多糖，谱图足够简单，正如讲义所说，一维谱有时就够了。分支会增加难度：分支点的残基在两个或更多位置上带取代基，每多一条支链就多一套重叠的环上信号，图中那个大结构就是这种情况。

要点 · Tip α/β 判断口诀（葡萄糖型）：J 大（7–8 Hz）是 β，J 小（3–4 Hz）是 α。

SLIDE 72 2D NMR of Carbohydrates: TOCSY, COSY, ROESY 糖类的二维 NMR：TOCSY、COSY、ROESY



IN ONE LINE · 一句话

TOCSY isolates each ring, COSY orders the protons within it, ROESY finds which rings are joined and where.

TOCSY 把每个环单独拎出来，COSY 排出环内氢的顺序，ROESY 找出哪些环相连、连在哪里。

TOCSY 全相关谱 · **COSY** 相关谱 · **ROESY** 旋转坐标系 NOE 谱 · **inter-residue NOE** 残基间 NOE

EN

Most carbohydrate problems need 2D spectra, and the three panels show a division of labour. Each sugar ring is a closed spin system, because there is no proton–proton coupling across the glycosidic oxygen. A TOCSY trace starting from a resolved anomeric proton (panel A, traces labelled by residue) therefore pulls out all the protons of that one ring from the crowded 3 to 4 ppm region. COSY (panel B) steps from

中文

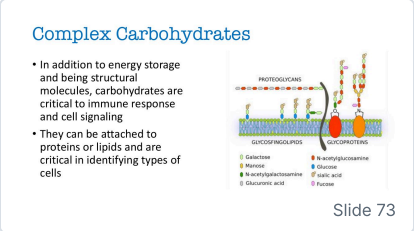
多数糖类问题需要二维谱，这三张小图展示了三种实验的分工。每个糖环是一个封闭的自旋体系，因为隔着糖苷键的氧没有氢–氢耦合。所以从一个分得开的异头氢出发的 TOCSY 迹线（图 A，按残基标注）能把这一个环上所有的氢从拥挤的 3–4 ppm 区域里提取出来。COSY（图 B）从邻居走到邻居，H1 → H2 → H3，把顺序定下

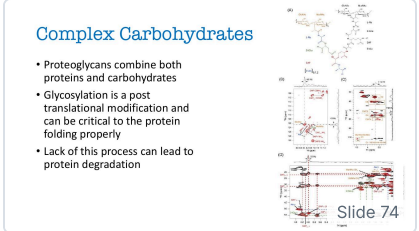
neighbour to neighbour, H1 to H2 to H3, which fixes the order. Neither experiment can cross the glycosidic bond. That requires a through-space experiment: in ROESY (panel C) the anomeric H1 of one residue shows a cross peak to the proton on the linkage carbon of the next residue, for example H1 of residue II to H4 of residue III, which gives both the sequence and the linkage position. ROESY is used in place of NOESY because oligosaccharides are of intermediate size, where the ordinary NOE passes through zero, while the ROE is always positive.

来。这两个实验都跨不过糖苷键，跨过去要靠通过空间的实验：在 ROESY（图 C）里，一个残基的异头氢 H1 与下一个残基连接碳上的氢之间有交叉峰，例如残基 II 的 H1 到残基 III 的 H4，这就同时给出了连接顺序和连接位置。之所以用 ROESY 而不用 NOESY，是因为寡糖属于中等大小的分子，普通 NOE 在这个范围内会经过零点，而 ROE 始终为正。

要点 · Tip 跨糖苷键的另一种办法是通过键的 HMBC：异头氢到对面连接碳的三键相关（H1–C1–O–Cx）。这是前面课上学过的实验在糖上的直接应用。

SLIDES 73–74 Glycoconjugates: Glycoproteins, Glycolipids, Proteoglycans 糖缀合物：糖蛋白、糖脂、蛋白聚糖





IN ONE LINE · 一句话

Sugars attached to proteins and lipids act as identity tags on cells and help proteins fold.

连在蛋白和脂质上的糖是细胞的身份标签，还帮助蛋白折叠。

- **glycoprotein** 糖蛋白
- **proteoglycan** 蛋白聚糖
- **glycosphingolipid** 鞘糖脂
- **glycosylation** 糖基化
- **post-translational modification (PTM)** 翻译后修饰

EN

Carbohydrates do more than store energy (starch, glycogen) and build structures (cellulose, chitin). The first figure shows the outer face of a cell membrane, covered with sugar chains attached to proteins (glycoproteins and proteoglycans) and to lipids (glycosphingolipids). The legend lists the monosaccharides involved: galactose, mannose, glucose, fucose, sialic acid, N-acetylglucosamine and others. These chains form the cell's molecular identity. The immune system reads them to tell self from foreign, and the ABO blood groups are differences in a single terminal sugar. Proteoglycans are proteins carrying long carbohydrate chains, often with more carbohydrate than protein by mass. Glycosylation, the enzymatic attachment of sugars after the protein has been synthesized, is a post-translational modification. Many proteins need it to fold correctly, and a protein that lacks its glycans may misfold and be degraded by the cell's quality control. For structure work, glycans are a major complication: they are flexible and heterogeneous, so they are often invisible in crystal structures, and the second figure shows the dense set of 2D NMR spectra needed to characterize one glycan.

中文

糖类的功能不止储能（淀粉、糖原）和搭建结构（纤维素、几丁质）。第一张图画的是细胞膜的外表面，上面覆盖着连在蛋白上的糖链（糖蛋白、蛋白聚糖）和连在脂质上的糖链（鞘糖脂）。图例列出了涉及的单糖：半乳糖、甘露糖、葡萄糖、岩藻糖、唾液酸、N-乙酰葡糖胺等。这些糖链构成细胞的分子身份。免疫系统靠它们区分「自己」和「外来」，ABO 血型的差别只是末端一个糖不同。蛋白聚糖是带有长糖链的蛋白，按质量算糖常常比蛋白还多。糖基化是蛋白合成之后由酶把糖接上去的过程，属于翻译后修饰。许多蛋白需要它才能正确折叠，缺少糖链的蛋白可能错误折叠，并被细胞的质量控制系统降解。对结构研究来说，糖链是个大麻烦：它们柔性大、不均一，在晶体结构里常常看不见；第二张图展示了表征一条糖链需要多少密密麻麻的二维 NMR 谱。

PART 8

Synthetic Polymers

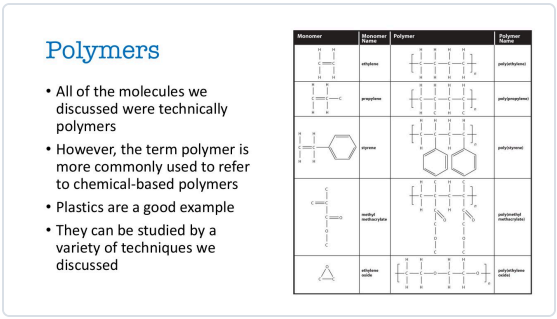
合成高分子

Slides 75–78

The same spectroscopic tools, pointed at plastics: chemistry by IR, stereochemistry and defects by NMR.

同样的波谱工具用在塑料上：红外看化学变化，NMR 看立体规整性和结构缺陷。

SLIDE 75 Polymers 高分子



Slide 75

IN ONE LINE · 一句话

Everything today was a polymer, but 'polymer' usually means synthetic ones such as plastics.

今天讲的都是聚合物，但「高分子」一词通常指塑料这类合成聚合物。

polymer 聚合物 / 高分子 · **repeat unit** 重复单元 · **polyethylene / polypropylene** 聚乙烯 / 聚丙烯 · **polystyrene** 聚苯乙烯 · **PMMA** 聚甲基丙烯酸甲酯

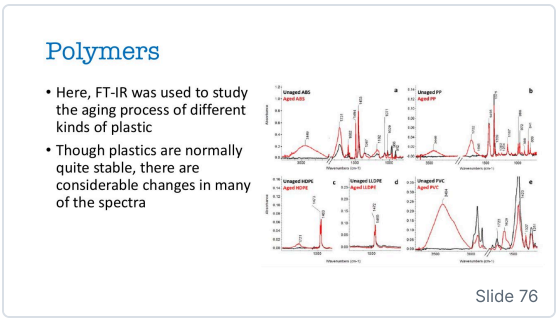
EN

Proteins, nucleic acids and polysaccharides are all polymers in the strict sense, but in everyday chemical usage the word refers to synthetic materials. The table on the slide pairs monomers with their polymers: ethylene gives polyethylene, propylene gives polypropylene, styrene gives polystyrene, methyl methacrylate gives PMMA and ethylene oxide gives poly(ethylene oxide). There is one important difference from biopolymers. A protein is a single defined molecule with an exact sequence and mass. A synthetic polymer sample is a population of chains with a distribution of lengths, and often with variations in stereochemistry and branching. Characterization therefore reports averages and distributions, and the structural questions are: what is the repeat unit, how regular is the chain, and what defects does it contain? IR, NMR and MS from earlier lectures all apply.

中文

蛋白质、核酸、多糖严格说都是聚合物，但在日常化学用语里，这个词指的是合成材料。讲义上的表把单体和聚合物一一对应：乙烯得聚乙烯，丙烯得聚丙烯，苯乙烯得聚苯乙烯，甲基丙烯酸甲酯得 PMMA，环氧乙烷得聚环氧乙烷。它们和生物高分子有一个重要区别。一个蛋白质是确定的单一分子，序列和质量都是精确的。一份合成高分子样品则是一群链，链长有分布，立体化学和支化程度也常有差异。所以表征给出的是平均值和分布，要回答的结构问题是：重复单元是什么？链有多规整？含有哪些缺陷？前面讲过的 IR、NMR、MS 都能用上。

SLIDE 76 FT-IR: Watching Plastics Age FT-IR：观察塑料老化



IN ONE LINE · 一句话

Aged plastics grow C=O and O–H bands: oxidation, visible directly in the spectrum.

老化的塑料长出 C=O 和 O–H 吸收带：氧化过程直接体现在谱图上。

FT-IR 傅里叶变换红外 · **carbonyl band** 羰基吸收带 · **photo-oxidation** 光氧化 · **hydroxyl band** 羟基吸收带

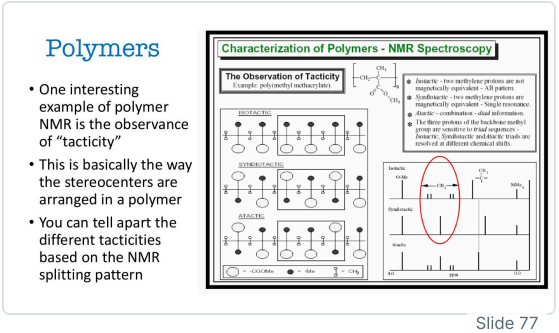
EN

The five panels overlay the IR spectra of unaged (black) and aged (red) samples of ABS, polypropylene, HDPE, LLDPE and PVC. Plastics are regarded as stable, yet the red traces differ clearly from the black ones. The diagnostic changes are the ones you would predict from the functional-group table. Polyethylene and polypropylene are pure hydrocarbons and should have no absorption between about 1650 and 1800 cm^{-1}. After ageing, a band appears near 1715 to 1735 cm^{-1}, the C=O stretch of ketones, acids and esters formed when UV light and oxygen attack the chain. A broad band near 3400 cm^{-1} (O–H) grows as well, most clearly in the PVC panel. IR is well suited to this job: it needs almost no sample preparation, works on solids, and reports directly on the functional groups that oxidation creates. The ratio of the carbonyl band to a C–H reference band is a standard measure of how degraded a plastic is.

中文

五张小图把未老化（黑）和老化（红）的 ABS、聚丙烯、HDPE、LLDPE、PVC 的红外谱叠在一起。人们通常认为塑料很稳定，但红线和黑线明显不同。有诊断意义的变化，正是按官能团表可以预测到的。聚乙烯和聚丙烯是纯烃，在大约 1650–1800 cm^{-1} 之间本不该有吸收；老化之后，1715–1735 cm^{-1} 附近出现了吸收带，这是紫外线和氧气进攻分子链后生成的酮、酸、酯的 C=O 伸缩振动。3400 cm^{-1} 附近的宽带（O–H）也在增长，PVC 那张图最明显。红外很适合做这件事：几乎不需要制样，固体可以直接测，而且直接反映氧化产生的官能团。羰基带与某个 C–H 参比带的强度比，是衡量塑料降解程度的常用指标。

SLIDE 77 NMR of Polymers: Tacticity 高分子 NMR：立构规整度



IN ONE LINE · 一句话

Tacticity = the pattern of stereocentres along the chain; the backbone CH_2 signal reveals it.

立构规整度 = 立体中心沿链的排列方式；主链 CH_2 的信号能把它显示出来。

tacticity 立构规整度 · **isotactic** 全同立构 · **syndiotactic** 间同立构 · **atactic** 无规立构 · **diastereotopic** 非对映异位 · **AB quartet** AB 四重峰

EN

In a polymer such as PMMA, every second backbone carbon is a stereocentre. Tacticity describes how neighbouring stereocentres are related. Isotactic: all substituents on the same side. Syndiotactic: strictly alternating. Atactic: random. Tacticity matters in practice, because it controls whether chains can pack and crystallize, and hence the melting point and strength of the material. The NMR test, circled in red on the slide, uses the backbone CH_2 protons and is an application of symmetry from earlier lectures. In an isotactic chain the two CH_2 protons are in different environments (one faces the ester groups on

中文

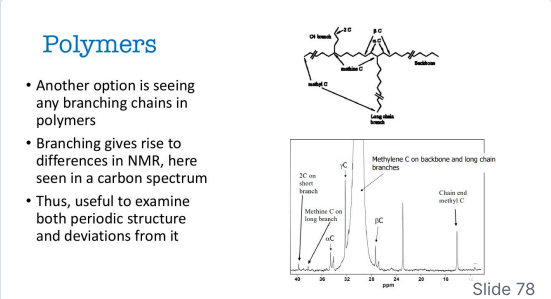
在 PMMA 这样的聚合物里，主链上每隔一个碳就是一个立体中心。立构规整度描述相邻立体中心之间的关系。全同立构：取代基都在同一侧。间同立构：严格交替。无规立构：随机排列。立构规整度有实际意义，因为它决定分子链能否紧密堆积和结晶，进而决定材料的熔点和强度。NMR 的判别方法（讲义上红圈处）用的是主链 CH_2 的氢，是前面讲过的对称性的应用。全同立构链里，CH_2 的两个氢环境不同（一个两侧都对着酯基，另一个两侧都对着甲基），它们是非对映异位的，位移不同，彼此耦合，给出四条线的 AB 图

both sides, the other faces the methyls). They are diastereotopic, have different shifts, couple to each other and give an AB pattern of four lines. In a syndiotactic chain a local twofold axis interchanges the two protons, so they are equivalent and give one singlet. An atactic sample contains both arrangements and shows a mixture. The α-methyl singlet goes further: it resolves three signals for the three possible triads of neighbouring stereocentres, so the proportions of each can be measured by integration.

间同立构链里，局部的二重轴使这两个氢互换，它们等价，给出一个单峰。无规立构样品两种排列都有，谱图是混合的。α-甲基的单峰提供更多信息：它能分辨出相邻立体中心的三种三单元组各自的信号，积分就能定量各自的比例。

要点 · Tip 记法：全同 → CH_2 两个氢不等价 → AB 四重峰；间同 → 等价 → 单峰。本质上是判断非对映异位氢，和小分子里的做法完全一样。

SLIDE 78 NMR of Polymers: Branching 高分子 NMR：支化



IN ONE LINE · 一句话

Branch points and chain ends are chemically distinct carbons, so ^{13}C NMR can identify and count them.

支化点和链端是化学环境不同的碳，^{13}C NMR 可以鉴别并计数。

branching 支化 · **methine carbon** 次甲基碳 · **chain end** 链端 · **α, β, γ carbons** α、β、γ 碳

EN

A perfectly linear polyethylene chain is one repeated CH_2 group and would give a single ^{13}C peak near 30 ppm, the large off-scale signal in the spectrum. Real polyethylene has branches, and every branch creates carbons in new environments: the methine CH at the branch point, the backbone carbons one, two and three bonds away (labelled α, β and γ), the carbons of the branch itself, and methyl groups at chain ends (near 14 ppm). The spectrum on the slide assigns these small peaks one by one, including separate signals for short branches and long branches. Their positions identify the type of branch, and their integrals relative to the main CH_2 peak give the number of branches per thousand carbons. This number has commercial meaning: low-density polyethylene (flexible film) is highly branched and high-density polyethylene (rigid containers) is nearly linear. The slide's closing sentence is a fair summary of polymer NMR. The large peaks give the periodic structure and the small ones give the deviations from it, and the deviations often determine the properties.

中文

完全线性的聚乙烯链只有一种重复的 CH_2，^{13}C 谱上只会在 30 ppm 附近出一个峰，也就是谱图里那个高得出界的信号。真实的聚乙烯有支链，每条支链都会带来环境不同的碳：支化点上的次甲基 CH，主链上距支化点一、二、三根键的碳（标为 α、β、γ），支链自身的碳，以及链端的甲基（14 ppm 附近）。讲义上的谱把这些小峰逐一归属了，短支链和长支链各有不同的信号。峰的位置说明支链的类型，相对于 CH_2 主峰的积分给出每一千个碳有多少条支链。这个数字有商业意义：低密度聚乙烯（柔软的薄膜）支化度高，高密度聚乙烯（硬质容器）几乎是线性的。讲义最后一句话是对高分子 NMR 的恰当总结：大峰给出周期性结构，小峰给出对周期性的偏离，而性能往往由这些偏离决定。

APPENDIX A

Choosing a Technique

技术对比总表

四种方法各自回答什么问题、需要什么样品、卡在哪里。这是本讲最值得背下来的一张表。

	X-ray crystallography X 射线晶体学	Cryo-EM 冷冻电镜	Solution NMR 溶液核磁	Tandem MS 串联质谱
Sample **样品**	Single crystal (or powder) 单晶（或粉末）	Solution, flash-frozen in vitreous ice 溶液，速冻成玻璃态冰	Solution, usually ^{13}C/^{15}N-labelled 溶液，通常需 ^{13}C/^{15}N 标记	Digested peptides, gas-phase ions 酶切肽段，气相离子
What is measured **测的是什么**	Diffraction from electrons → electron density map 电子的衍射 → 电子密度图	Projection images → 3D potential map 投影图像 → 三维势图	Chemical shifts, J couplings, NOEs 化学位移、J 耦合、NOE	Masses of fragment ions 碎片离子的质量
Structural level **结构层次**	Secondary to quaternary 二级到四级	Secondary to quaternary 二级到四级	All levels, plus dynamics and binding 各级结构，外加动态与结合	Primary only 只有一级
Size range **适用大小**	No upper limit if it crystallizes 能结晶就没有上限	Best for large particles and complexes 最适合大颗粒和复合物	Up to about 100 kDa (slide 41) 约 100 kDa 以内（讲义第 41 页）	Any, after digestion 酶切后不限
Main strength **主要优势**	Highest resolution; very detailed 分辨率最高，细节最丰富	No crystal; near-native state; several conformations 不需晶体；接近天然状态；可分出多种构象	No crystal; real solution state; disordered proteins 不需晶体；真实溶液状态；可做无序蛋白	Fast, sensitive identification 快速、灵敏的鉴定
Main limitation **主要局限**	Must crystallize; one conformation; fails on amorphous or disordered samples 必须结晶；单一构象；非晶和无序样品无效	Low signal per image; needs many particles; costly 单张图信号弱；需要大量颗粒；昂贵	Size limit; overlap; relaxation; isotope cost 分子量上限；重叠；弛豫；同位素昂贵	No fold information; depends on databases 没有折叠信息；依赖数据库
Hydrogens **氢原子**	Usually invisible 通常看不见	Better in principle (slide 24) 原理上更有利（第 24 页）	The nucleus actually observed 直接观测的就是氢核	Not applicable 不适用

APPENDIX B

Triple-Resonance Cheat Sheet

三共振实验速查

所有实验都以第 i 个残基的 N–H 为观测点。「看到的碳」一栏说明条带里出现的是哪个残基的哪种碳。相对灵敏度读自讲义第 61 页的柱状图，是近似值（HNCO = 100）。

Experiment 实验	Carbons seen from N–H of residue i 从第 i 个残基 N–H 看到的碳	Peaks per strip 每条峰数	Relative sensitivity 相对灵敏度	Role 用途
CBCANNH (= CBCANH, HNCACB)	Cα(i), Cβ(i) strong Cα(i−1), Cβ(i−1) weak 自己的强，前一个的弱	4	≈ 2–4 (i) ≈ 1–2 (i−1)	Own and previous residue; Cα and Cβ opposite sign 自己和前一个残基；Cα、Cβ 符号相反
CBCA(CO)NNH (= CBCA(CO)NH)	Cα(i−1), Cβ(i−1) only 只有前一个残基	2	≈ 9–13	Identifies the previous residue without ambiguity 无歧义地确定前一个残基
HNCA	Cα(i) strong, Cα(i−1) weak	2	≈ 50 (i) ≈ 15 (i−1)	Cα-only walk; more sensitive 只用 Cα 行走；灵敏度更高
HN(CO)CA	Cα(i−1) only	1	≈ 71	Partner of HNCA HNCA 的搭档
HN(CA)CO	C′(i) and C′(i−1)	2	≈ 13 (i) ≈ 4 (i−1)	Carbonyl walk 用羰基碳行走
HNCO	C′(i−1) only	1	100	Partner of HN(CA)CO; the most sensitive 3D experiment HN(CA)CO 的搭档；最灵敏的三维实验
HCCH-TOCSY	All ^{1}H of one side chain 同一侧链的全部氢	—	—	Which protons belong to the side chain 哪些氢属于这条侧链
HCCH-COSY	^{1}H on the neighbouring carbon only 只有相邻碳上的氢	—	—	Order along the side chain 侧链上的先后顺序
^{15}N NOESY-HSQC	All ^{1}H within ~5 Å of the N–H N–H 周围约 5 Å 内的所有氢	many	—	Distance restraints → tertiary structure 距离约束 → 三级结构

Reading the names · 名字怎么读

Letters list the nuclei in the order the magnetization visits them. A nucleus in parentheses is passed through without recording its shift. "(CO)" in a name means the route crosses the carbonyl, and that makes the experiment see only residue i−1.

字母按磁化经过原子核的顺序排列。括号里的核只是路过，不记录位移。名字里带 “(CO)” 表示路径经过羰基，这样的实验只能看到第 i−1 个残基。

Typical ^{13}C shifts · 典型 ^{13}C 位移

C′ ≈ 175 ppm · Cα 50–65 ppm · Cβ 15–45 ppm.
Gly: Cα ≈ 45, no Cβ. Ala: Cβ ≈ 19. Ser: Cβ ≈ 64. Thr: Cβ ≈ 70.

甘氨酸没有 Cβ；丙氨酸的 Cβ 最低；丝氨酸、苏氨酸的 Cβ 比自己的 Cα 还高。这四种残基是把片段对到序列上的路标。

APPENDIX C

Worked Example: Sequential Assignment

例题：主链顺序归属

Four HSQC peaks, labelled A to D in no particular order, give the strips below. The shifts are invented for practice, but the method is exactly that of slides 47 to 52. Work out the order of the four residues and their types.

四个 HSQC 峰，随意标为 A 到 D，它们的条带数据如下。位移数值是为练习编的，方法与讲义第 47–52 页完全相同。求这四个残基的先后顺序和类型。

HSQC peak 峰	CBCANNH, strong pair 强的一对	CBCANNH, weak pair 弱的一对	CBCA(CO)NNH
A	62.1 (Cα), 69.8 (Cβ)	58.4, 63.9	58.4, 63.9
B	52.5 (Cα), 19.1 (Cβ)	45.2 only	45.2 only
C	45.2 (Cα), no Cβ	56.0, 30.1	56.0, 30.1
D	58.4 (Cα), 63.9 (Cβ)	52.5, 19.1	52.5, 19.1

1 **Separate own from previous. 分清自己和前一个。**

The peaks that also appear in CBCA(CO)NNH belong to residue i−1. In every row they coincide with the weak pair, so the strong pair is the residue's own Cα and Cβ.

在 CBCA(CO)NNH 里也出现的峰属于第 i−1 个残基。每一行里它们都和弱的一对重合，所以强的一对就是这个残基自己的 Cα、Cβ。

2 **Match previous to own. 用「前一个」去找「自己」。**

A's previous residue has 58.4/63.9, which are D's own shifts, so D comes just before A. D's previous is 52.5/19.1 = B's own, so B precedes D. B's previous is 45.2 with no Cβ = C's own, so C precedes B.

A 的前一个残基是 58.4/63.9，正是 D 自己的位移，所以 D 紧挨在 A 前面。D 的前一个是 52.5/19.1，等于 B 自己的，所以 B 在 D 前面。B 的前一个是 45.2 且没有 Cβ，等于 C 自己的，所以 C 在 B 前面。

3 **Write the chain. 写出顺序。**

(56.0 / 30.1) → **C** → **B** → **D** → **A**

4 **Identify residue types. 判断残基类型。**

C: Cα 45.2 and no Cβ → Gly. B: Cβ 19.1 → Ala. D: Cβ 63.9, above its Cα → Ser. A: Cβ 69.8 → Thr.

C：Cα 45.2 且无 Cβ → 甘氨酸。B：Cβ 19.1 → 丙氨酸。D：Cβ 63.9，比自己的 Cα 还高 → 丝氨酸。A：Cβ 69.8 → 苏氨酸。

5 **Place the fragment on the sequence. 对到序列上。**

The fragment is X–Gly–Ala–Ser–Thr. Find where G-A-S-T occurs in the protein's known sequence. The residue before Gly has Cα 56.0 and Cβ 30.1; several residue types fit those values (for example Glu, Gln, Lys, Arg), so the sequence decides which one it is.

这段是 X–Gly–Ala–Ser–Thr。到蛋白的已知序列里找 G-A-S-T 出现的位置。甘氨酸前面那个残基的 Cα 为 56.0、Cβ 为 30.1，符合这组数值的残基类型不止一种（例如 Glu、Gln、Lys、Arg），具体是哪个由序列决定。

要点 · Tip 如果链中出现脯氨酸，它没有 N–H，自己没有条带，行走在这里中断。但它后面那个残基的条带里仍然能看到脯氨酸的 Cα/Cβ，所以可以知道「前面是脯氨酸」，然后从另一段重新开始走。

APPENDIX D

Which Method for Which Molecule

各类大分子用什么方法

Class 类别	Monomer and linkage 单体与连接	Primary structure from 一级结构靠什么	3D structure and detail from 三维结构与细节靠什么	Special difficulty 特有的难点
Proteins 蛋白质	20 amino acids, peptide bond 20 种氨基酸，肽键	Tandem MS; NMR backbone assignment 串联质谱；NMR 主链归属	XRD, cryo-EM, NMR (NOESY)	Membrane proteins; disordered proteins; size limit in NMR 膜蛋白；无序蛋白；NMR 的分子量上限
Nucleic acids 核酸	4 nucleotides, phosphodiester 4 种核苷酸，磷酸二酯键	Sanger sequencing 桑格测序	XRD (fibre and crystal), NMR (imino NOESY, ^{31}P)	Only four similar monomers, so heavy overlap 只有四种相似的单体，重叠严重
Carbohydrates 糖类	Monosaccharides, glycosidic bond 单糖，糖苷键	NMR for everything: anomeric ^{1}H and ^{3}J for α/β; TOCSY and COSY within a ring; ROESY or HMBC across the linkage 全部靠 NMR：异头氢和 ^{3}J 定 α/β；TOCSY、COSY 定环内；ROESY 或 HMBC 跨糖苷键		Isomers of equal mass; linkage position; α/β; branching 质量相同的异构体；连接位置；α/β；分支
Synthetic polymers 合成高分子	One or a few monomers 一种或几种单体	Known from the synthesis 由合成方法决定	FT-IR (functional groups, ageing); ^{1}H NMR (tacticity); ^{13}C NMR (branching) FT-IR（官能团、老化）；^{1}H NMR（立构）；^{13}C NMR（支化）	A mixture of chain lengths, so results are averages 链长有分布，结果是平均值

APPENDIX E

Glossary

术语表

按英文字母排序，共 225 条；右栏是该术语首次出现的讲义页码。

English	中文	Slide
2D classification	二维分类	21–22
θ–2θ geometry	θ–2θ 几何	6
3D classification	三维分类	18
3D NOESY-HSQC	三维 NOESY-HSQC	57
AB quartet	AB 四重峰	77
air scatter	空气散射	12
α/β barrel (TIM barrel)	α/β 桶	30–31
α, β, γ carbons	α、β、γ 碳	78
α / β configuration	α / β 构型	70
α-helix	α 螺旋	27
α-synuclein	α-突触核蛋白	62–63
amide	酰胺	43
amorphous	非晶态 / 无定形	5
anomeric carbon	异头碳	70
anomeric proton	异头氢	71
anti conformation	反式构象	68
anti-scatter slit	防散射狭缝	12
atactic	无规立构	77
axial / equatorial divergence	轴向 / 赤道发散	8
backbone assignment	主链归属	47
background	背景	11
base pair	碱基对	64
beam overflow	光束溢出	9–10
β-hairpin	β 发夹	29
β-sheet	β 折叠	27
biopolymer	生物高分子	26
branching	分支	71
Cα / Cβ	α 碳 / β 碳	45
capillary gel electrophoresis	毛细管凝胶电泳	65–66
carbonyl band	羰基吸收带	76
carbonyl carbon (C′)	羰基碳	45
carbonyl relay	经羰基中继	50
chain end	链端	78
chain termination	链终止	65–66
characteristic X-rays	特征 X 射线	7
chemical shift anisotropy (CSA)	化学位移各向异性	58–59
chemical shift dispersion	化学位移分散度	40
class	类	21–22
class average	类平均	20
coiled coil	卷曲螺旋	29
collimate	准直	12
collision cell	碰撞室	36–37
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APPENDIX F

Self-Test

自测题

共 16 题，覆盖全部八个部分。先自己答，再翻到后面对答案。

1. A Cu Kα diffractometer (λ = 1.5406 Å) shows a peak at 2θ = 30.0°. What is d (n = 1)?
   铜靶衍射仪（λ = 1.5406 Å）在 2θ = 30.0° 处有一个峰，求晶面间距 d（n = 1）。
2. Why does a glass slide give a broad hump and no sharp peaks?
   玻璃片为什么只有一个宽包，没有尖峰？
3. A powder sample is packed slightly below the correct height. What happens to the peaks?
   粉末样品装得比正确高度略低，峰会怎样？
4. In cryo-EM, why is the grid plunged into liquid ethane?
   冷冻电镜为什么要把载网插进液态乙烷？
5. You average 4 times as many particle images. By what factor does the signal-to-noise ratio improve?
   参与平均的颗粒图像数增加到 4 倍，信噪比提高多少倍？
6. Which level of protein structure does tandem MS give, and why is it weak for a new protein?
   串联质谱给出蛋白质的哪一级结构？为什么对新蛋白不好用？
7. Why is the protein HSQC a ^{1}H–^{15}N experiment, and which residues give no backbone peak?
   蛋白质的 HSQC 为什么用 ^{1}H–^{15}N？哪些残基没有主链峰？
8. A CBCANNH strip shows four peaks. How do you decide which two belong to the previous residue?
   CBCANNH 的一条里有四个峰，怎么判断哪两个属于前一个残基？
9. A residue's own strip shows a single carbon peak at 45 ppm and no Cβ. Which residue is it?
   某残基自己的条带里只有一个 45 ppm 的碳峰，没有 Cβ，它是什么残基？
10. Why is HNCO the most sensitive triple-resonance experiment?
    为什么 HNCO 是最灵敏的三共振实验？
11. When does TROSY help, and when does it hurt?
    TROSY 什么时候有用，什么时候反而吃亏？
12. Which NMR experiment gives tertiary-structure information, and on what physical basis?
    哪个 NMR 实验给出三级结构信息？物理基础是什么？
13. A ^{1}H–^{15}N HSQC has all its peaks between 8.0 and 8.5 ppm in ^{1}H. What do you conclude?
    一张 ^{1}H–^{15}N HSQC 的峰在氢维全部落在 8.0–8.5 ppm，说明什么？
14. Why does a ddNTP stop DNA synthesis?
    ddNTP 为什么能终止 DNA 合成？
15. A glucose residue shows an anomeric doublet with J = 7.8 Hz. α or β? Which 2D experiment connects it to the next ring?
    某葡萄糖残基的异头氢是 J = 7.8 Hz 的双峰，是 α 还是 β？哪个二维实验能把它和下一个环连起来？
16. The backbone CH_2 of a PMMA sample is a clean singlet. What is the tacticity? And aged polyethylene shows a new IR band at 1715 cm^{-1}: what happened?
    某 PMMA 样品的主链 CH_2 是一个干净的单峰，它是什么立构？老化的聚乙烯在 1715 cm^{-1} 出现新的红外吸收，发生了什么？

Answers

答案

1. θ = 15.0°, sin θ = 0.2588, d = 1.5406 / (2 × 0.2588) = 2.98 Å.
   先把 2θ 除以 2。θ = 15.0°，d ≈ 2.98 Å。

2. Glass is amorphous. With no long-range periodic spacing, no single d satisfies Bragg's law over many planes; only a broad distribution of near-neighbour distances scatters.
   玻璃是非晶态，没有长程周期性，不存在能在大量晶面上同时满足布拉格条件的 d，只有近邻距离的宽分布在散射。

3. They shift to lower 2θ and become weaker and broader (slide 11), so every calculated d is too large.
   峰移向低 2θ，并且变弱变宽（讲义第 11 页），算出来的 d 全都偏大。

4. To cool so fast that water vitrifies. Ice crystals would damage the protein and diffract electrons strongly.
   为了快到让水玻璃化。冰晶会破坏蛋白，还会强烈衍射电子。

5. By √4 = 2.
   √4 = 2 倍。

6. Primary structure only. Identification relies on matching fragment spectra to a database of known sequences, and de novo reading is not exact.
   只有一级结构。鉴定靠碎片谱与已知序列数据库比对，从头测序不够准确。

7. Each residue has one backbone N–H, giving one peak per residue, and ^{15}N provides the dispersion. Proline (no amide H) and the N-terminal residue give no peak.
   每个残基有一个主链 N–H，所以每个残基一个峰，^{15}N 提供分散度。脯氨酸（没有酰胺氢）和 N 端残基不出峰。

8. They are the weaker pair, and they are the two that also appear at the same N–H position in CBCA(CO)NNH. The second test is the reliable one.
   它们是较弱的一对，而且在 CBCA(CO)NNH 的同一 N–H 位置上也出现。后一条判据更可靠。

9. Glycine.
   甘氨酸。

10. It has the fewest transfer steps, and they use the large, efficient N–C′ coupling, so little signal is lost to relaxation.
    它的传递步数最少，而且走的是较大、高效的 N–C′ 耦合，因弛豫损失的信号少。

11. It helps for large proteins at high field (best near 900 MHz for ^{1}H–^{15}N). For small proteins or low fields it only discards signal.
    对大蛋白、在高场下有用（^{1}H–^{15}N 最佳约 900 MHz）。对小蛋白或低场，它只是白白丢掉信号。

12. NOESY. Cross-relaxation through space depends on $1/r^6$, so cross peaks mark protons within about 5 Å.
    NOESY。通过空间的交叉弛豫正比于 $1/r^6$，交叉峰标出相距约 5 Å 以内的氢。

13. The protein is disordered or unfolded: every amide experiences a similar averaged environment.
    蛋白是无序的或没有折叠：所有酰胺感受到的都是相近的平均环境。

14. It has no 3′-OH, so no phosphodiester bond can be formed to the next nucleotide.
    它没有 3′-OH，无法与下一个核苷酸形成磷酸二酯键。

15. β (H1 and H2 both axial). ROESY or NOESY through space, or HMBC through bonds; COSY and TOCSY cannot cross the glycosidic oxygen.
    β（H1、H2 都是直立氢）。通过空间用 ROESY 或 NOESY，通过键用 HMBC；COSY 和 TOCSY 跨不过糖苷键的氧。

16. Syndiotactic (the two protons are equivalent). The polyethylene has been oxidized, forming C=O groups.
    间同立构（两个氢等价）。聚乙烯被氧化，生成了 C=O。

APPENDIX G

Points to Check in the Slides

讲义中需要留意的地方

下面汇总正文里所有「留意」提示。它们是讲义措辞与通行说法不完全一致、或容易读错的地方。考试以授课老师的说法为准，有疑问建议直接向老师确认。

位置	说明
Slide 5	讲义写的是 “intermolecular spacings” 和 “internuclear distances”。严格说，X 射线是被电子散射的，直接得到的是电子密度图（讲义第 39 页也这么说）；原子核位置是从电子密度的中心推出来的。
Slide 7	讲义文字写的是 “tungsten anode emits electrons ⋯ strike the cathode, often cobalt or copper”，电极名称与通常的定义相反。标准说法：发射电子的是阴极（钨灯丝，cathode），被轰击并发出 X 射线的是阳极靶（Cu、Co、Mo 等，anode）。同一页的示意图里 anode disk 也是被轰击的那一方。考试若照讲义原句出题，建议向老师确认以哪个说法为准。
Slide 12	讲义首句 “One the sample⋯” 应为 “Once the sample⋯”，笔误，不影响理解。
Slides 34–35	讲义写 “embedded in the wall”，这里的 wall 指的是细胞膜（membrane）。细胞壁（cell wall）是另一种结构，不要混淆。
Slide 48	讲义最后一条 “It gives the preceding Ca and Cb” 容易读成「只给出前一个」。完整的说法是：它既给出本残基的，也给出前一个残基的。只给前一个的是下一页的 CBCA(CO)NNH。
Slide 49	① 图中注释说丝氨酸、苏氨酸因 Cβ 的 “low chemical shift” 而容易辨认。按数值看，它们的 Cβ 位移其实很高（约 64 和 70 ppm），属于低场（downfield）；这张谱的 ^{13}C 轴是向下增大的，所以它们出现在图的下方。这里的 “low” 应理解为低场或图上位置靠下，而不是 ppm 数值小。② 强弱只是经验规律，偶尔两对峰强度相近，这时无法单凭强度判断哪对属于前一个残基。下一个实验正是为了消除这种歧义。
Slide 51	讲义写 “gives you the next amide proton in the chain”。按实验给出的信息，行走方向是朝 i−1（朝 N 端）；倒过来问「谁的条带把我当作前一个残基」，就能朝 C 端走。两个方向都可以，关键是分清条带里哪对峰属于自己。
Slide 64	讲义写 “Made of 4 base pairs (GTCA or GUCA)”。准确地说是 4 种碱基（bases）；碱基对（base pairs）指配对关系，只有两类：A 配 T（RNA 中为 U），G 配 C。